

Philosophy and Mathematics.

A Propaedeutic Treatise

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The author of the present treatise, who neither is nor wishes to be a mathematician, has found in philosophy, the general theory of science, an incitement to learn the language of mathematics as well, and, with respect to its leading fundamental concepts, so far as possible, to enter into its way of thinking.

I.

Mathematical Functions.

The word “function” occurs in a very wide sense. Wherever there can be talk of acting, suffering, doing, function denotes a particular law-governed activity. It is said of an official¹, that he is in function; one speaks of the function of the eye, of the function of a nerve, and so forth.

On the domain of pure magnitudes there can, as regards these, just as little be talk of real doing as of real suffering: an a cannot act, a b cannot suffer, an x cannot function in the likeness either of an official or of a nerve.

Since nevertheless the designation “function” is not only retained, but even held fast as a concept in mathematics, one must be able, in the mathematical application, to point out something corresponding to the fundamental meaning.

To be sure, pure magnitudes cannot exert real activity, but by being placed in different connections they can nevertheless essentially alter their mutual relations. A magnitude can, so to speak, be appointed to different offices; it can, as an addend ($a+x$), as a factor (ax), as an exponent of a power (a^x , x^a), as a logarithm, and so forth, acquire a very different influence, and to that extent appear in different functions.

Here, however, is another side of the concept, which must be brought out especially in the theory of magnitude. One speaks not merely of functioning, of being in function; one speaks also of that which is brought about by the functioning activity. Outside mathematics one thinks, in connection with function, chiefly of the activity, in mathematics chiefly of the result; outside mathematics the point is to be in function at something, within mathematics to be a function of something. The law-bound relation between a dependent and an independent shows itself, however, on both sides.

¹With categorical emphasis Chamberlain Norden in “Gamle Minder” stresses his “term of function.”

“If a magnitude y depends, according to a certain law, on other magnitudes $x, z, t, u \dots$, so that every value of y is determined by the values arbitrarily assigned to $x, z, t, u \dots$, then y is said to be a function of these magnitudes, all regarded as variable, y dependent, the others independent. This is commonly denoted: $y = f(x, z, t, u \dots)$ ”².

According to this mathematical determination of the concept, changes will thus proceed from the independents and exert their lawful influence upon the dependent. In order to recall the likeness between non-mathematical and mathematical functions, one might indeed regard the independents $x, z, t, u \dots$ as the functioning ones, the agents, that cause the changes in the suffering y . But since the conception is figurative, one likewise sees why $x, z, t, u \dots$ are not said to be in function with respect to y , whereas y is expressly said to be a function of $x, z, t, u \dots$.

In the function, then, the magnitudes are variable; but magnitudes can of course also be invariable. That every magnitude must be equal to itself, and that every magnitude can, by increase or decrease, at every moment become unequal to itself, is a contradiction that is resolved in the concept of function. In and for itself no magnitude is either invariable or variable, for no magnitude is anything alone in and for itself. Whether the particular magnitude, on the other hand, in its relation to other magnitudes, is to play the role of the invariable or of the variable, and, within variability, again that of the dependent or of the independent, is determined by the function, and in turn determines the function.

If the concept function is now, in the first place, determined by the concept of dependence, then it depends again on determining the mode of dependence, that is to say: on pointing out the law.

In the mathematical presentation of the different functions there shows itself a scale of case, rule, and law, which must become an object of philosophy’s attention.

$$\text{That } 2(3 + 4) = 2 \cdot 3 + 2 \cdot 4 = 6 + 8 = 2 \cdot 7 = 14,$$

$$\text{that } (3 \cdot 4)^2 = 12^2 = 3^2 \cdot 4^2 = 9 \cdot 16 = 144,$$

$$\text{that } 2^3 \cdot 2^4 = (2 \cdot 2 \cdot 2) \cdot (2 \cdot 2 \cdot 2 \cdot 2) = 2^{3+4} = 2^7,$$

can be experienced by trial, and seen by comparison. In this way one can test various

²C. Ramus. *Algebra og Functionslære*. Kjøbenhavn 1840, S. 14.

cases and abstract a rule from them. But, however important the rule may be for practical use, it is nevertheless not, like the law, a necessary expression of the essence of the matter.

In order to obtain a rule as comprehensive as possible, one must test as many cases as possible; but even if one tested cases by the thousand without meeting a single exception, the rule could not thereby be raised to a law.

The rule expresses only *that a relation is*, not that it necessarily must be; but the law holds as law in virtue of the concept. That, when the law is to be known, it does not depend on the number of cases, but on *the essence of the matter*, and hence on insight, can be seen from the following example.

That $\frac{1}{1-x} = 1 + x + x^2 + \dots$ will yield an infinite series is proved not by continuing the division to infinity, but by understanding what the division, on the given presupposition, necessarily brings with it.

How easily a series of cases falling under a rule can deceive, when one would from this infer a law, is well known. One seeks, for example, a law for the prime numbers, and finds: $2^3 - 1 = 7$, $2^5 - 1 = 31$, $2^7 - 1 = 127$, all prime numbers. But if one now, from these three cases, would draw the conclusion that $2^{2m-1} - 1$ must necessarily be a prime number, one would badly miscalculate, since already the fourth case, $2^9 - 1 = 511 = 7 \cdot 73$, gives no prime number.

That in the cited example the exception to the rule already occurs with the fourth case is, in the whole consideration, itself as it were a matter of chance. Had the exception occurred with the 99th case, we should still just as little have had any law, but should on the contrary have been more exposed to being deceived by the multitude of cases³.

That mathematics, instead of definite numerical values, employs general designations — instead of

$$\begin{array}{ll} 2 \cdot (3 + 4) = 2 \cdot 3 + 2 \cdot 4 & a(x + y) = ax + ay; \text{ instead of} \\ (3 \cdot 4)^2 = 3^2 \cdot 4^2 & (xy)^a = x^a y^a; \text{ instead of} \\ 2^3 \cdot 2^4 = 2^{3+4} & a^x a^y = a^{x+y} \end{array}$$

³Fermat supposed that $2^{(2^n)} + 1$ was a prime number; but Euler showed that $2^{32} + 1$ is divisible by 641.

— is, as regards the cognition of laws, an important step. The understanding, which from the beginning was entangled in the consideration of the definite values, the diverse cases, the individual rules, raises itself, by means of the general designations, to insight into the essence of the relations, and thereby to insight into the law.

How intimately the law hangs together with the concept can be seen by a glance at the logarithmic function. $a^z = x$, $z = \log x$; $a^t = y$, $t = \log y$, cannot be established by any proof. Here there can be talk neither of case nor of rule nor of law; here it is a question of a determination of the concept for the logarithm, a statement of its category. But once the category of the logarithm is fixed, then there holds as law in virtue of the concept: $a^z a^t = a^{z+t} = xy$, and $z + t = \log(xy) = \log x + \log y$.

In so far as the concept of function everywhere points to definite relations of dependence, we are directed by the functions to perform intellectual, law-governed actions. How intimately these actions fuse with the concept can be seen from the so-called fundamental equations.

As the first of all functions, $a+x$ cannot indeed itself be defined by any fundamental equation; but the four single functions ax , x^a , a^x , and $\log x$ can, on the other hand, be regarded as entirely determined in their nature⁴ by the following four equations.

$$f_1(x+y) = f_1(x) + f_1(y); \quad f_1(x) = ax, \quad \text{for } a(x+y) = ax + ay.$$

$$f_2(xy) = f_2(x) f_2(y); \quad f_2(x) = x^a, \quad \text{for } (xy)^a = x^a y^a.$$

$$f_3(x+y) = f_3(x) f_3(y); \quad f_3(x) = a^x, \quad \text{for } a^{x+y} = a^x a^y.$$

$$f_4(xy) = f_4(x) + f_4(y); \quad f_4(x) = a \log x, \quad \text{for } a \log(xy) = a \log x + a \log y.$$

If we now see in these fundamental equations a demonstration of laws, a presentation of concepts, the difference between mathematics and philosophy will be conspicuous. What is peculiar makes itself known in the language.

For a philosophical presentation of the concept the word is used, and the word sounds different in the different vernaculars; but mathematics is not content with the word; independent of the diversity of the vernaculars, the exact science has its own language: $f(x+y) = f(x) f(y)$ is neither Danish nor French, but mathematical.

⁴Jvfr. Ramus: Algebra og Functionslære S. 14–18.

That the notation which mathematics employs is now also really a language in a stricter sense than, for example, the notation of notes in music, is seen from the fact that it is not confined to indicating different operations of calculation, but expresses oppositions, draws conclusions, and points out the difference between the possible and the impossible.

In the inverse functions we have expressions both for opposition and for conclusion. If the function φ is the inverse of ψ , i.e. $\varphi(\psi(x)) = x$, then also $\psi(\varphi(x)) = x$, i.e. the function ψ is the inverse of φ . If $\varphi(x) = x + a$, then $\psi(x) = a - x$; if $\varphi(x) = ax$, then $\psi(x) = \frac{x}{a}$; if $\varphi(x) = x^n$, then $\psi(x) = \sqrt[n]{x}$; if $\varphi(x) = a^x$, then $\psi(x) = \log x$; if $\varphi(x) = \cos x$, then $\psi(x) = \arccos(x)$.

The last designation, which indicates a relation between the trigonometric line $\cos x$ and the arc corresponding to it, $\arccos(x)$, expresses the opposition in such a way that φ and ψ do not signify two mutually cancelling actions; for x is opposed to x in that in $\varphi(x)$ it is an arc, in $\psi(x)$ a straight line.

That mathematics, in its language, can express both the merely possible and the in-itself impossible, shines forth from the manner in which the relation between the rational and the irrational, the real and the imaginary magnitudes is indicated: $a = \frac{b}{c}$ is, for thought, a reality, $b = \pm \sqrt{a^2 - x^2}$ a possibility, and $\sqrt{-1}$, $\sqrt[2m]{-a}$ an impossibility.

While philosophy, then, together with the other sciences⁵, is referred to a grammatical vernacular, mathematics avails itself essentially of a pasigraphic sign-language: this difference of language must surely have its deeper ground. In so far as a language expresses thoughts and concepts, it expresses how the universal and the individual are separated in abstractions and united in judgments, distinguished in judgments and connected in conclusions.

In the grammatical language the universal has a one-sided preponderance over the individual; in the mathematical language the universal is determined so punctually that it can coincide with the single and meet the individual. In the grammatical language the concept assumes a logical form; in the pasigraphic sign-language a mathematical form.

Logic says: "A is A", "every thing is what it is", and thereby shows that in the

⁵Incomplete sign-systems, such as, for example, the chemical, cannot be placed on the same level as the mathematical.

single it looks only at the universal.

Mathematics posits $A = A$, and thereby shows that in the universal it expressly looks at the single.

In the logical proposition “ A is A ”, A is like an image that represents “every thing”, but in itself is without significance. In the designation of the single, one does not have this single, but only the universal: the word and the thing, the designation and the designated fall apart from one another.

In the mathematical proposition $A = A$, A also indeed represents any magnitude whatsoever, but, inasmuch as A is an example of any magnitude whatsoever, B , C , D ..., it, as a magnitude, at the same time represents itself; $A = A$, for $2 = 2$, $3 = 3$: the universal is posited punctually, the sign and the designated are congruent.

Logic transforms the equation $A = A$ into a grammatical proposition; mathematics transforms the proposition A is A into an equation. Already in this is seen an essential difference between the exact and the dialectical science.

In the proposition as equation there is no difference between subject and predicate. This holds not only of the so-called identical equations, such as $A = A$, but of any equation whatsoever. For what stands on the one side of the equals-sign represents the subject, and what stands on the other side, the predicate; but for every equation it is quite indifferent from which side it is read. If $f(x + y) = f(x) + f(y)$, then of course also $f(x) + f(y) = f(x + y)$. In the mathematical language the predicate is congruent with the subject, for mathematics is an exact science.

If, on the other hand, the equation is transformed, in the grammatical language, into a logical proposition, then an essential difference arises between subject and predicate; for which reason these two components cannot indeed be united by the mere equals-sign, but must be joined by the copula. Now, when the difference between the subject and the predicate in the logical judgment is brought to consciousness, there appears the much-discussed contradiction, that the single is said to be the universal⁶. In the logical language the predicate’s being one with the subject is a contradiction,

⁶Already the Megarian Stilpo attacked the logical judgments. He maintained that one could say nothing but identical, or rather tautological, propositions, for example “Man is man,” “Horse is horse”; but not “The horse runs,” for “horse” and “to run” are not identical; the latter predicate one ascribes also to dogs and lions. Jvfr. Paul Møllers *Forelæsninger over den ældre Philosophies Historie*.

and philosophy a dialectical science.

The mathematical sign-language and the grammatical proposition-language are, in their origin, both natural to sound human understanding, and therefore enter into various combinations.

Mathematical determinations of relation are so far at home in existence that daily life consumes a by no means insignificant quantum of conscious and unconscious mathematics for practical need. Expressions such as “heavy, heavier, quick, quicker, warm, warmer” still fall outside the mathematical; but, when it is expressly asked: “how heavy?” “how much heavier?” and so forth, the answer can scarcely lack a mathematical element.

The peculiar relation of mathematical thinking to the general-logical shows itself perhaps most strikingly when mathematics itself takes the word in order to pose a problem. Example. “A water-reservoir can be filled by the uniform inflow of several pipes, n in number, and the times in which it is filled by each pipe separately are given: $a_1, a_2, a_3 \dots a_n$. In what time (x) will it be filled, when the inflow takes place through all of them simultaneously⁷?”

He who has not learned mathematics will understand what is here said, and yet not understand it. In its logical fundamental features the matter is not only universally intelligible, but can even become so clear that the problem is solved in the same moment it is posed.

For if one has four pipes of such a kind that a uniform inflow through each of them separately can fill a certain water-reservoir in one hour, then sound human understanding naturally grasps, involuntarily, that the same water-reservoir must, by uniform inflow through all four pipes at once, be filled in a time four times as short, that is, in the course of a quarter of an hour. Here the mathematical merges into the general-logical.

But how different this mathematical is nevertheless from the general-logical is also evident from the cited example. One need only make the relation between the times in which the water-reservoir is filled through each pipe separately a little complicated; and at once the problem is so peculiar that the stiffest logician, the most

⁷C. Ramus. *Elementær Algebra* S. 179.

acute dialectician, who does not know mathematics, racks his brains in vain to solve it.

He who understands the mathematical language finds the problem easy, indeed for every case essentially equally easy; for the equation $\frac{x}{a_1} + \frac{x}{a_2} + \frac{x}{a_3} + \dots \frac{x}{a_n} = 1$, which gives $x = \frac{1}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots \frac{1}{a_n}}$, fits all possible cases. This merging of the universal into the individual cases is decisive for mathematical cognition.

In the logical language the formula is a conclusion, in the mathematical the conclusion is a formula. In the formal conclusion the validity of the concept is indeed seen, but always in such a way that the single and the particular vanish into the universal. In the formula the universal is punctualized, and thereby fitted for the exact determination of the particular and the single.

From the well-known syllogism:

“All men are mortal,
Cajus is a man.
Therefore: Cajus is mortal.”

one obtains, although Cajus is expressly named, nevertheless no particular information about Cajus as Cajus. In the minor premise “Cajus is a man” the universal has already so far swallowed up the single that one gains absolutely nothing by inserting into the syllogism either Brutus or Pompejus or Cnejus instead of Cajus.

From the formula $x = \frac{1}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots \frac{1}{a_n}}$, on the other hand, one obtains, by inserting for $a_1, a_2, a_3 \dots$ a multitude of arbitrary values, the most diverse cases. And yet the formula is universally valid, invariable; only the individual cases are variable. By the formula, then, there is brought about a peculiar agreement between the universal and the single, the invariable and the variable.

With the logical formula-business, at the present stage of the sciences, things stand but poorly; with the mathematical formulas the matter has another bearing. Against the logical formula-business there is a scientific remedy: a sound and sharp dialectic; against the mathematical formula-business no remedy is needed, for insight into the real laws is conditioned by insight into the mathematical formulas⁸.

⁸By its formulas mathematics is just fitted to express the real laws. If, in the determination of

The scientific character in the development of these formulas can already be seen from the manner in which the concept of function is articulated in the several fundamental functions.

The single functions $a \log x$, a^x , x^a , ax , $a + x$ can, namely, be derived in such a way that the succeeding one springs with inner necessity from the preceding. However much difference there may be here between these functions in their independent appearance, one nevertheless easily finds, on the thread of development, the knots where the one lies enclosed in the other.

Here, thus, is a turning-point, where the concept of multiplication dawns beneath the veil of addition. If one takes $a + a + a$, a sum of three equal addends, one comes almost unawares to conceive the magnitude as an a repeated 3 times. But therewith the transition is not yet made. Consciousness can often come very near to a new concept, can glimpse it as in a dawn, and yet the concept can again vanish and be lost. If a new concept is to arise for cognition, it is not enough that the investigation leads up to the turning-point where it has its origin (for then the great discoveries of the present would surely have been made centuries ago); in order that a new concept may come to be in cognition, it is required that thought rightly catch sight of it, take it up into a category, introduce it among the other concepts, follow its course, and test its consequences.

As it goes in mathematics with the transition from addition to multiplication, so also with the transition from multiplication to the exponential function, and from this to the logarithm⁹.

In the equation $a + a + a + \dots a = na$ the same magnitude is conceived on the one side as sum, on the other side as product. Here too it is expressly seen that the

these laws, it merely came to naming the universal, they could be determined without the help of mathematics; but since it also comes to the phenomena and the individual cases, inasmuch as the laws are the laws of the phenomena, the mathematical must come in, that the expression may become exact. He who, for example, grasps the law of uniformly increasing velocity only in so far as it is logically expressed in words, grasps indeed in a certain sense the essential, but, since he lacks the punctuality in the particular cases, not the real. — It makes a great difference whether the law for the collision of hard bodies is set forth in logical form or in mathematical formula. The logical presentation in words is, with all its accuracy, always hovering; only a formula such as $\frac{MH+mh}{M+m}$ is fitted to explain the single.

⁹The inner connection of the trigonometric functions with the exponential and with the logarithm can only be pointed out later.

mathematical notation is a language.

As an immediate expression of calculation the equation indicates only that on each side of the equals-sign there are equal magnitudes; but with respect to the concept the equation is a judgment.

Seen from the logical side, this judgment again shows the contradiction, the contradiction that multiplication is thus both the same and yet not the same as addition. This contradiction, together with the *diversus respectus* by which formal logic would avoid the contradiction, and just thereby comes into contradiction with itself, and falls under dialectics, does not concern mathematics at all.

As a mathematical judgment the equation will not logically declare what multiplication either is or is not, but rather demonstrate, in the exact form, how the intellectual action of multiplication is connected with that of addition. What dialectics is in philosophy, the intellectual action with its practical transitions is in mathematics. That the matter stands thus, we can illustrate by a retrospect.

The mathematical function presupposes the variable in its dependents as in its independents; only by grasping the infinite possibility of the changes does one grasp the function. But the function presupposes, in all these changes, at the same time an invariable; it expresses, after all, the law, and the law is the invariable.

Seen from the finite side, it might well seem as if the invariable in the formula were a dead schema, indifferent toward the variable cases — that, for example, with respect to the formula $x = \frac{1}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots \frac{1}{a_n}}$, it could be quite indifferent what values one inserted. This is indeed, in the finite sense, so. But the more finite the point of view becomes, the more the concept of function also recedes into the background; the more, on the contrary, the concept of function prevails, the more intimately is seen the unity of the variable and the invariable: the function points precisely to that which dialectics demands.

In the above problem the times $a_1, a_2 \dots a_n$ were given, each for itself, and consequently independent of one another. But one might assume a_1 given in such a way that $a_2, a_3 \dots$ were made dependent on it, and that according to such a law that they formed, for example, either an arithmetic or a geometric series: $a_1 = a_1$, $a_2 = a_1 + d$, $a_3 = a_1 + 2d \dots$ or $a_1 = a_1$, $a_2 = a_1 q$, $a_3 = a_1 q^2, \dots$ thus in the first case

$a_p = a_1 + (p-1)d = \varphi_1(p)$; in the second case $a_p = a_1 q^{p-1} = \varphi_2(p)$; we then obtained

$$x = \frac{1}{\frac{1}{\varphi(1)} + \frac{1}{\varphi(2)} + \dots + \frac{1}{\varphi(p)} + \dots + \frac{1}{\varphi(n)}} = F(\varphi(p)) = f(p).$$

Since in the expression for x there is found in the denominator a sum of the form $\frac{1}{\varphi(p)}$, which is a function of $\varphi(p)$, and the whole sum consequently a function of $\varphi(p)$, so also 1, divided by the sum, is a function of $\varphi(p)$. The invariable and the variable will in this way interpenetrate one another.

The mathematical functions are not dead formulas, but active, busy, practical concepts that lead an intellectual life in the operations of science. For such practical transitions no dialectical spur is needed.

As an example of how the mathematical concepts of action can cooperate even within elementary limits, we will, in conclusion, recall the symmetric functions.

Through

$$\begin{aligned} (x - a_1)(x - a_2) &= x^2 - (a_1 + a_2)x + a_1 a_2 = x^2 + A_1 x + A_2; \\ (x - a_1)(x - a_2)(x - a_3) &= x^3 - (a_1 + a_2 + a_3)x^2 + (a_1 a_2 + a_1 a_3 + a_2 a_3)x - a_1 a_2 a_3 \\ &= x^3 + A_1 x^2 + A_2 x + A_3, \end{aligned}$$

one will understand the general expression:

$$f(x) = x^n + A_1 x^{n-1} + A_2 x^{n-2} + A_3 x^{n-3} + \dots + A_{n-1} x + A_n = (x - a_1)(x - a_2)(x - a_3) \dots (x - a_n) = 0.$$

Here the invariable is everywhere present in the variable; for the law of development shows us the reciprocal determination of coefficients and roots:

$A_1 = -(a_1 + a_2 + a_3 + \dots + a_n) = f_1(a_1 a_2 \dots a_n)$ = the sum of all the a 's with opposite sign; here the first fundamental function operates.

$A_3 = -(a_1 a_2 a_3 + a_1 a_2 a_4 + \dots + a_1 a_2 a_n + a_2 a_3 a_4 + \dots + a_{n-2} a_{n-1} a_n) = f_3(a_1 a_2 a_3 \dots a_n)$ = the sum of the products of the a 's taken three at a time, with opposite sign; here the first and second fundamental functions cooperate. $A_n = (-1)^n (a_1 a_2 a_3 \dots a_n) = f_n(a_1, a_2, \dots, a_n)$ = the product of all the a 's with opposite sign, when n is odd. It is thus

the second fundamental function that here appears.

If one considers besides that the series proceeds according to powers of x , whereby the alternation of the exponential function with the logarithmic also becomes possible, then one obtains a total impression of the intellectual cooperation of the functions.

The formal concepts of logic are like empty vessels, which must either remain in their emptiness or be filled with matter and content from without; the function-concepts of mathematics are like active organs that assimilate to themselves an essential content.

Although the mathematical operations, in their peculiar procedure, belong to the specialized science without concerning philosophy, their character nevertheless lends itself to a philosophical consideration, to illuminate the activity of thought from a side of its own, and to make visible problems whose treatment, not to say solution, is and must be the aim of philosophy.

II.

The Problem of Infinity.

If the concept of function can serve to determine the unity of the invariable with the variable, then it must also be able to serve to clear up the relation between the finite and the infinite. *How does the finite stand to the infinite?*

The problem of infinity is the great common problem of mathematics and philosophy. But, before one can rightly think of the solution of a problem, one must first of all see to letting the problem come to be. The present problem is now of such a kind that it can stand clear before consciousness only when mathematics will help philosophy to set it forth.

With interest in the problem, we then follow the way marked out by mathematics.

1) Series and Trigonometric Functions.

The usual representation, that the infinite falls outside the finite, and can be reached only by an approach, seems at once to find its confirmation in the series-expansions of elementary mathematics.

The concept of series is a further development of the concept of function; for each term in the series is a function of its place, the expression for the general term $u_n = f(n)$ shows that u_n is dependent on n . According as now $n = 0$, or $= 1$, $= 2$, $= 3 \dots$ or $n = -1$, $= -2$, $= -3 \dots$, $u_n = u_0$, $= u_1$, $= u_2$, $= u_3 \dots$ or $u_n = u_{-1}$, u_{-2} , $u_{-3} \dots$ will change its value.

Since now n , by being diminished or increased by a unit, can be changed to infinity, the same must hold of u_n . The series $\dots u_{-2}, u_{-1}, u_0, u_1, u_2, \dots$ then represents a *progressus* and *regressus in infinitum*, whose separate terms are finite, but their number infinite.

To the three kinds of continued proportion, the arithmetic, the harmonic, and the geometric, there corresponds, as is well known, a typical difference between the following series:

α) The arithmetic series (the equidifference series), which presupposes $u_n = f_1(n) = a + nx$:

$$\dots a - 3x, a - 2x, a, a + x, a + 2x, a + 3x \dots$$

β) The harmonic series, according to which $u_n = f_2(n) = \frac{1}{a + nx}$:

$$\dots \frac{1}{a - 3x}, \frac{1}{a - 2x}, \frac{1}{a - x}, \frac{1}{a}, \frac{1}{a + x}, \frac{1}{a + 2x}, \frac{1}{a + 3x} \dots$$

γ) The geometric series (the equiquotient series), which is determined by $u_n = f_3(n) = ax^n$:

$$\dots \frac{a}{x^3}, \frac{a}{x^2}, \frac{a}{x}, a, ax, ax^2, ax^3 \dots$$

In these series the finite strives by different ways toward the infinite, but infinity shows itself partly as an abrupt annihilation of the finite, partly as the unattainable.

In the arithmetic series the approach on the right side of a proceeds toward ∞ , while on the left side there can, when nx becomes $= a$, occur a case in which a single term vanishes.

In the harmonic series the development proceeds to the right, from $\frac{1}{a}$, steadily onward toward 0; while on the contrary to the left the case can occur in which ∞ suddenly swallows up a single term.

In the geometric series the development advances equally to both sides, to the right (with $x > 1$) toward ∞ , to the left toward 0.

So conceived, ∞ and 0 are then absolute negations, mere limits of magnitude, but not themselves magnitudes. Formal logic accordingly holds fast to the principle that the finite is not infinite, and the infinite not finite. But however incontestable these negative judgments, guaranteed by the principle of contradiction, may seem to be, so poor, so empty of content are the corresponding positive ones: “the finite is the finite, the infinite the infinite”, for their so-called identity is a pure tautology.

As a dialectical science, philosophy has not been able to remain standing at such tautologies. That the infinite is not finite means, in other words, that it is a limit for the finite, thus the Not-Finite. The finite, which is the Not-Infinite, is then at the same time a limit for the infinite. But when the finite and the infinite are in this way each other’s limits, the determination of the limit must become dialectical.

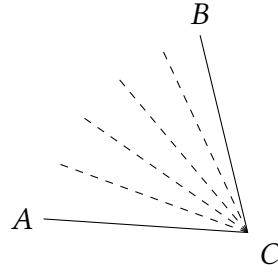
For if the infinite has its limit in the finite, then it has a finitude-limit, a finite limit, and is itself finite. If, conversely, the finite first reaches its limit in the infinite, then it has no finite limit and is consequently infinite.

Under such a tension between formal logic and dialectics, it is of importance to investigate on which side mathematics will place itself.

It might at first glance seem as if mathematics would unconditionally maintain the principle: every thing is what it is, $a = a$, $0 = 0$, $\infty = \infty$, and thereby settle the matter. But if now dialectics added: every thing is incessantly in the process of being what it is not, even 0 can according to its concept be as ∞ and ∞ conversely as 0; would mathematics then, when it rightly bethinks itself, contradict this?

In order to grasp mathematics’ real meaning, we must call to mind the difference, emphasized in the previous section, between the equation and the logical judgment. What the logical judgment is for dialectics, the equation is for the operation.

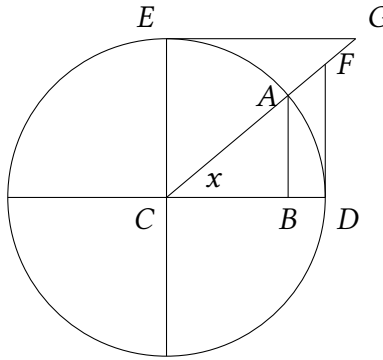
As regards the mathematical judgment, we must not listen too much to the literal expressions in spoken language, but above all look at the factual development of concepts in the sign-language. If we begin with an ideal facticity, and ask, for example, how many angles the finite angle ACB may be said to enclose, then mathematics answers in plain words: “infinitely many, infinitely small”. The



infinite is thus, both as the infinitely great and the infinitely small, enclosed by finite limits. But if the infinite is really enclosed by finite limits, then it must surely, says sound understanding, in one way or another be itself finite.

How does this now really hang together? As guiding examples we may for once consider the trigonometric functions.

$$AB = f_1(x) = \sin x; \quad BC = f_2(x) = \cos x; \quad DF = f_3(x) = \operatorname{tg} x, \quad EG = f_4(x) = \operatorname{cot} x \text{ etc.}$$

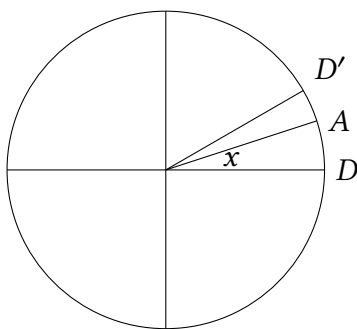


As x , our independent variable, grows in the first quadrant, there grow with it, as is well known, $\sin$, tg , $\sec$, whereas $\cos$, cotg , cosec ... decrease. The changes in x proceed, in the first quadrant, from 0° to 90° ; it is necessary to consider these changes more closely.

If $x = 0$, then $\sin x = 0$, but $\cos x = 1$; $\operatorname{tg} x = 0$, but $\cot x = \infty$, $\sec x = 1$, but $\operatorname{cosec} x = \infty$; if $x = 90^\circ$, then $\cos x = 0$, $\sin x = 1$, $\operatorname{tg} = \infty$, $\cot x = 0$, etc.

The finite and the infinite are here determined by one another in such a way that the transition from finite to finite coincides with the transition from finite to infinite, whereby then also a transition from finite to 0 coincides with a transition from finite to ∞ .

What does the expression $\angle x = 0$ mean? In the finite sense it might mean that here there was no x at all; but in virtue of the concept of function $\angle x = 0$ means a Not-Being which is nevertheless a Being. That an angle $= 0$ is no angle goes without saying; that $\angle x = 0$ is nevertheless regarded as being is seen from the fact that a cosine can only be the cosine of an angle, and that $\cos 0$ is just as surely a real magnitude as $\cos 1^\circ$.



If A is assumed to be at a distance of one degree from D and D' , then it follows that the way from 1° to 0° is not longer than the way from 1° to 2° . A can come as near to D as one wills; so long as A does not coincide with D , $\angle x$ is a finite magnitude; but the moment A coincides with D , $\angle x$ becomes $= 0$. So long as x is still finite, $\cot x$ is also finite; but the moment x becomes $= 0$, $\cot x$ becomes $= \infty$.

The distance from finite to 0 and from finite to ∞ is thus here like a distance from the one point of finitude to the other. If the one point of finitude lies infinitely near to the other, then the same can now be said of 0 and finite, of ∞ and finite.

If, in order to illuminate the relation between the series and the trigonometric lines, one would fetch an analogy from aesthetics, one would have to say: the trigonometric functions are classical, the infinite series romantic.

To have the infinite enclosed by the finite is classical; to conceive the infinite as something beyond the finite, unattainable, romantic. Which of these ways of viewing does mathematics now support?

In so far as mathematics is an exact science, one would suppose that it held to the classical; but if we look at its use of categories such as “approach, infinite approach”, one would almost believe that it was enthusiastic for the romantic.

With this hint it is naturally not the intention to want to mix the aesthetic into the

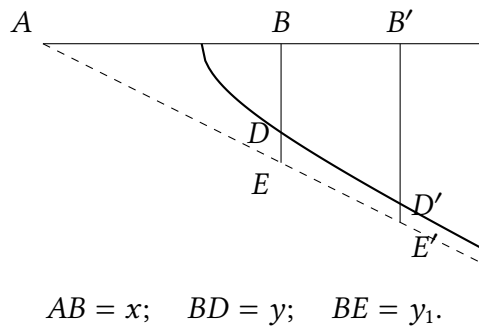
mathematical, or to clear up scientific concepts by unscientific analogies. The fleeting allusion to the aesthetic should merely serve to bring out the difference between the twofold manner in which the relation of the finite to the infinite is conceived.

What, then, lies in the concept: approach, infinite approach?

By “infinite approach” one cannot, in the first place, think of an approach toward an Attainable; for the Attainable expresses precisely that the approach must end with the attainment of the goal, and thus be in its essence finite. An approach toward an Attainable contains no contradiction; an Unattainable without approach is likewise no contradiction. There is thus nothing contradictory in the proposition that parallel lines, however far they be extended, will never intersect one another. When mathematics sometimes uses the expression, of parallel lines, that they, extended to infinity, will intersect one another: then one cannot really thereby think of any approach; it is an impossibility here presented under the form of possibility.

An infinite approach, an approach toward an Unattainable, on the other hand, seems to present the hardest of all contradictions. By an approach one must, after all, in one sense or another come nearer to the goal; but to come nearer to the goal must surely be the same as to come nearer to the attainment; but from the attainment of the unattainable one is always equally far away.

None the less, mathematics maintains the concept of an infinite approach, and beats down every contradiction by incontestable proofs. So in the doctrine of asymptotes.



For the hyperbola one has, for example, the equation:

$$f(x) = y = \frac{b}{a} \sqrt{x^2 - a^2};$$

for the hyperbola's asymptote: $f_1(x) = y_1 = \frac{b}{a}x$. Accordingly,

$$\begin{aligned} y_1 - y &= \frac{b}{a} (x - \sqrt{x^2 - a^2}) = \frac{b}{a} \cdot \frac{(x - \sqrt{x^2 - a^2})(x + \sqrt{x^2 - a^2})}{x + \sqrt{x^2 - a^2}} \\ &= \frac{b}{a} \cdot \frac{a^2}{x + \sqrt{x^2 - a^2}}. \end{aligned}$$

The consideration of this formula shows, 1) that here there is a real approach, for the greater x becomes, the smaller naturally becomes the fraction $\frac{a^2}{x + \sqrt{x^2 - a^2}}$; 2) that the goal of the approach is absolutely unattainable, for the fraction is 0 only when x has become $= \infty$.

What validity, then, may we ascribe to the concept of approach?

2) Series with Increasing Powers.

If the concepts "Unattainable" and "approach" are to be brought into logical connection, then one might, according to "*diversus respectus*", perhaps begin by thinking of the matter thus: the category "approach" must always be viewed from two standpoints: 1) the point from which one sets out, the point of departure, and 2) the point that is to be reached, the goal, the end-point.

If there is any movement at all from a definite point of departure toward a goal, then one may, if the goal is infinitely far away, indeed in one sense grant that every approach is as a nothing against the infinite remoteness of the goal, and yet, when one looks back, regard the continued removal from the point of departure as an approach to the goal. Here, then, in one respect — namely in so far as one looks forward — there is no approach, but yet in another respect — when, be it noted, one looks back — a real approach.

Dialectics could now, to be sure, in this connection very easily confound the "*diversus respectus*", and elicit from formal logic the confession that it, according to its own explanation, thus after all sees the approach only as — removal, sees forward only in so far as it sees backward, and that just this is a contradiction; but we have placed ourselves before the forum of mathematics, and dare not anticipate its verdict.

The inexhaustible possibility of the changes, which lies hidden in the concept

of function as in its ground, steps by the intellectual action out of this ground, and becomes then, under given conditions, developed in the infinite series.

If a function of x is to be developed, within the limits of finitude, in a series according to powers, it is natural to begin with the highest power;

$$\begin{aligned} f(x) &= x^n + A_1x^{n-1} + A_2x^{n-2} + A_3x^{n-3} + \dots A_{n-1}x + A_n \\ &= (x - a_1)(x - a_2)(x - a_3) \dots (x - a_n) = 0, \end{aligned}$$

a series with descending powers is then a paradigm.

But when the task is to show how the finite, in infinite approach, strives toward the infinite, the paradigm must be changed, and the function developed according to increasing powers:

$$f(x) = A_0 + A_1x + A_2x^2 + A_3x^3 + \dots A_nx^m \dots$$

How such a series can present the infinite approach both to ∞ and, in its separate terms, to 0, is already clear from the geometric progression.

In the series $a \frac{1}{1-x} = a + ax + ax^2 + \dots ax^{n-1} \dots$, $x^\infty = \infty$ if $x > 1$. The series will then not only be capable of being continued through an infinite number of terms, but with each succeeding term come ever nearer to ∞ . For $x < 1$, $x^\infty = 0$. The series will thus, by being continued through countless terms, not only give a greater and greater sum, but each succeeding term become smaller and smaller, and thus approach 0.

Mathematics cannot, like formal logic, set itself at rest with an explanation of “infinite approach” in words; it uses its sign-language, and articulates the concept, in that for every infinite series it finds out the relation between the separate terms.

In order that this may happen, the concept of function must assume a definite character: $f(x)$, for example, $= a^x$. One then has the series $a^x = A_0 + A_1x + A_2x^2 + A_3x^3 + \dots A_mx^m \dots$

For the possible case $a^x = a^0$, it is easy to determine the series $a^0 = A_0$; the rest drops away. But $a^0 = A_0 = 1$. Since now $f(x) = a^x$ is to hold, according to the concept of function, for all possible values of x , one has in this function the first coefficient $A_0 = 1$ determined once for all.

Cartesius's method¹⁰ and Newton's binomial formula¹¹ give the mathematician, besides, means, as regards the remaining terms, to bring the variable to merge into the invariable, and, so to speak, to catch the infinite possibilities in the net of the relations.

Thus, in the first place, each coefficient becomes recognizable by the corresponding power. If, to keep to the given example, the power-factors $x^m z^n$ have in the one series the coefficients $A_m A_n$, in the other $A_{m+n} \frac{(m+n)(m+n-1)\dots(m+1)}{1 \cdot 2 \cdot 3 \dots n}$ ¹², then the mathematician has his equation:

$$A_{m+n} \frac{(m+n)(m+n-1)\dots(m+1)}{1 \cdot 2 \dots n} = A_m A_n$$

¹³ and is assured that it does not deceive. If, then, the mathematician has now on the whole secured for himself the relation between the powers and their coefficients, then he uses this relation again to find out thereby the relation between the coefficients among themselves.

In the cited series, thus, the whole swarm of possible coefficients is brought into a

¹⁰C. Ramus. *Algbr. F. S.* 82. — The method of undetermined coefficients. If $f(x) = A_0 + A_1 x + A_2 x^2 + \dots = B_0 + B_1 x + B_2 x^2 + \dots$, then A_0 must be $= B_0$, $A_1 = B_1$, $A_2 = B_2$, and so forth. Proof. For $x = 0$, $A_0 = B_0$ alone remains; by subtracting $A_0 = B_0$, one obtains $A_1 x + A_2 x^2 = B_1 x + B_2 x^2 \dots$, by dividing by x , $A_1 + A_2 x \dots = B_1 + B_2 x$, and, by setting $x = 0$, $A_1 = B_1$, and so on.

¹¹The binomial formula. From elementary algebra one knows: $(a+b)^2 = a^2 + 2ab + b^2$; $(a+b)^3 = a^3 + 3a^2 b + 3ab^2 + b^3 \dots$ and so forth. *Ram. Elmt. Algbr. S.* 35. In the higher mathematics (Ramus *Algbr. Functl. S.* 85) one presents $(p+x)^n = p^n + \frac{n}{1} p^{n-1} x + \frac{n(n-1)}{1 \cdot 2} p^{n-2} x^2 + \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3} p^{n-3} x^3 \dots$, a series according to descending powers of p and increasing powers of x , whose coefficients are formed, as the analogy shows, in a manner corresponding to the exponents of the powers.

¹²For one develops, namely, 1) two series according to the same law: $a^x = 1 + A_1 x + A_2 x^2 + \dots A_m x^m \dots$ and $a^z = 1 + A_1 z + A_2 z^2 + \dots + A_n z^n$, and obtains 2), by multiplying these two equations, two different but concordant developments: $a^x a^z = 1 + \dots A_m A_n x^m z^n \dots$ and $a^{x+z} = 1 + A_1(x+z) + A_2(x+z)^2 + \dots A_{m+n}(x+z)^{m+n} \dots$ From the cited term $A_{m+n}(x+z)^{m+n}$ there is developed again 3) the infinite series: $A_{m+n} \left[x^{m+n} + \frac{m+n}{1} x^{m+n-1} z + \frac{(m+n)(m+n-1)}{1 \cdot 2} x^{m+n-2} z^2 + \dots \dots \frac{(m+n)(m+n-1)(m+n-2)\dots(m+1)}{1 \cdot 2 \cdot 3 \dots n} x^m z^n \dots \right]$ 4) The particular term $A_{m+n} \frac{(m+n)(m+n-1)\dots(m+1)}{1 \cdot 2 \dots n} = A_m A_n$.

¹³The importance of the method of undetermined coefficients one will at once, among other things, appreciate from this, that the same magnitude can, without altering its value, present itself not merely under many, but under infinitely many forms. Among pure magnitudes too there take place — not, indeed, intellectual intrigues and mystifications, but yet so many changes of dress, disguises, and maskings, that he who does not see through the transformations is easily deceived. How easily, for example, may a beginner be dazzled by the many terms in the formula $\frac{n(n-1)(n-2)\dots(n-n+2)(n-n+1)(n-n)}{1 \cdot 2 \cdot 3 \dots n(n+2)(n-n+1)}$, so that he does not at once see the masked zero.

definite relation to a single one, namely A_1 , by the formula $A_m = \frac{A_1^m}{1 \cdot 2 \cdot 3 \dots m}$ ¹⁴; the last, sole unknown, A_1 , further brought into a definite relation to a by the formula: $a^{\frac{1}{A_1}} = 1 + \frac{1}{1} + \frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 2 \cdot 3} \dots$; this demonstrably converging sum being reckoned out, and the number found, $e = 2,718281828459 \dots$, adopted as the base in the natural, Neperian, logarithmic system.

By the relation between the exponential and the logarithmic function the coefficient A_1 is at last determined in the formula $A_1 = \frac{\log a}{\log e}$ ¹⁵, a relation which naturally leads again into new relations.

Instead of a logical consideration of the infinite approach in the infinite series, mathematics thus develops totalities of universally valid and yet peculiar, infinite and yet definite relations.

How, then, is the relation between the relations and the infinite determined?

3) The Systematics of the Functions.

The problem of infinity is a comprehensive problem of totality; if such a problem is to be solvable, it must, in the course of continued development, itself bring with it the conditions of the solution; these conditions must be more closely investigated.

That the infinite, to recall once more the dialectical, has its limit in the finite means, then, that it has its determinacy in the finite; that the finite at the same time has its limit in the infinite must accordingly mean that the determinations of finitude, by their countlessness, transgress every finite limit.

In the number-relation the finite is now, to be sure, most immediately determined by the finite; but with the infinite number-relations infinity itself at last merges into

¹⁴If one sets: $A_m A_n = A_{m+n} \frac{(m+n)(m+n-1) \dots (m+1)}{1 \cdot 2 \cdot 3 \dots n}$, and $n = 1$, one gets: $A_m A_1 = A_{m+1} \frac{m+1}{1}$, i.e. $A_{m+1} = \frac{A_m A_1}{m+1}$. If one next sets $m = 1$, one obtains $A_{1+1} = \frac{A_1 A_1}{1+1} = A_2 = \frac{A_1^2}{1 \cdot 2}$; if one sets $m = 2$, then one obtains $A_{2+1} = \frac{A_2 A_1}{2+1} = A_3 = \frac{A_2 A_1}{3} = \frac{\frac{A_1^2}{1 \cdot 2} A_1}{3} = \frac{A_1^3}{1 \cdot 2 \cdot 3} \dots A_m = \frac{A_1^m}{1 \cdot 2 \cdot 3 \dots m}$; from this: $a^x = 1 = A_1 x + A_2 x^2 + A_3 x^3 \dots = 1 + \frac{A_1 x}{1} + \frac{A_1^2 x^2}{1 \cdot 2} + \frac{A_1^3 x^3}{1 \cdot 2 \cdot 3} + \dots$ and from this again, when one sets $x = \frac{1}{A_1}$; $a^{\frac{1}{A_1}} = 1 + \frac{A_1}{1} \left(\frac{1}{A_1}\right) + \frac{A_1^2}{1 \cdot 2} \left(\frac{1}{A_1}\right)^2 + \frac{A_1^3}{1 \cdot 2 \cdot 3} \left(\frac{1}{A_1}\right)^3 + \dots = 1 + \frac{1}{1} + \frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 2 \cdot 3} = e$.

¹⁵If $a^{\frac{1}{A_1}} = e$, then $\frac{1}{A_1} \log a = \log e$, and $\log a = \log e A_1$; thus: $A_1 = \frac{\log a}{\log e}$ (log here denoting the logarithm in an arbitrary system).

the finite. The unity of the relations and the infinite is at this stage to be conceived in the concept “inner infinity”; but the concept “inner infinity” coincides again with the concept of function; the content of infinity can therefore first unfold itself in the systematic cooperation of the diverse functions.

For the elucidation of the systematics of the functions there must be emphasized a) that their development can be traced back to one fundamental formula, b) that their reciprocal relation and mutual transitions point to a common origin, c) that the one function can, in a continuous series, be developed out of the other.

a) The Fundamental Formula of the Functions.

If one were to attempt to enumerate the possible modes of calculation, rules, and formulas that occur in mathematical practice, one would soon receive an impression of a confusing, overwhelming multiplicity, of a boundless finitude that does not master the infinite. The more, on the contrary, the outwardly manifold is traced back to the first, general presuppositions, the more the concept prevails, and the more surveyable the whole becomes.

As a proof of the systematic procedure of mathematics there serves, then, this: that it not only reduces the countless derived functions to the simplicity of the fundamental functions, but even refers the development of all functions, elementary as well as derived, to one fundamental formula. On the condition that one sets, when required, for example $x + h$ instead of x , then

$$f(x) = A_0 + A_1x + A_2x^2 + A_3x^3 + \dots\dots$$

is a comprehensive fundamental formula.

Under the comprehensive fundamental formula the individuality of the several functions is as it were veiled; but it is only the finite that is here dissolved, in order to be grounded in the infinite. With the development the peculiar comes forth again, and the heterogeneous shows itself. On different conditions the common possibility becomes, namely, something different, and each function as it steps forth now sets its own conditions.

For $f(x) = a + x = A_0 + A_1x + A_2x^2 + \dots$, A_0 must become $= a$, $A_1 = 1$, $A_2, A_3 \dots = 0$.

For $f(x) = ax = A_0 + A_1x + A_2x^2 + \dots$, A_0 must become $= 0$, $A_1 = a$, $A_2, A_3 \dots = 0$.

For $f(x) = x^a = A_0 + A_1x + A_2x^2 \dots$, likewise only a single term acquires significance if x is monomial; if x is binomial, for example $= y + z$, the series reproduces the binomial formula.

For $f(x) = a^x = A_0 + A_1x + A_2x^2 \dots$ the definite expression is already given (S. 28).

For $f(x) = \log x = A_0 + A_1x + A_2x^2 \dots$ ¹⁶ a series-expansion cannot take place if x is monomial. Since $\log 0 = \pm\infty$, then, if x were 0, the first term A_0 would already be $= \infty$, and annihilate the significance of the series. If, on the other hand, one sets $1 + x$ instead of x , one obtains:

$$\log(1 + x) = \log e \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \right)$$

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If the trigonometric functions $\cos x$ and $\sin x$ are here added, and it is recalled that $\cos(-x) = \cos x$, and $\sin(-x) = -\sin x$, ($\cos 0 = 1$ and $\sin 0 = 0$), then the peculiarity will at once show itself, that $\cos x$ can contain only even, and $\sin x$ only odd powers

¹⁶If the fractional exponents are taken into consideration, various alterations again enter in a^x and $\log x$.

¹⁷Jvfr. Ramus. Algebr. og Funct. S. 87–88; further elucidation under b.

of x . One then obtains:¹⁸

$$\cos x = 1 + A_2x^2 + A_4x^4 + A_6x^6 \dots$$

$$\sin x = A_1x + A_3x^3 + A_5x^5 \dots$$

The peculiarity with which these functions step forth from the fundamental formula naturally also makes itself known in the result of the development:

$$\cos x = 1 - \frac{x^2}{1 \cdot 2} + \frac{x^4}{1 \cdot 2 \cdot 3 \cdot 4} - \frac{x^6}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} \dots$$

$$\sin x = x - \frac{x^3}{1 \cdot 2 \cdot 3} + \frac{x^5}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} - \frac{x^7}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7} \dots$$

When one fundamental law thus divides itself into different laws of development, of which each can again be regarded as a separate fundamental law for new laws of development, each of these again for new ones, and so forth, there is seen in the connection the systematic, and in the systematic connection the manner in which the finite relations become determinations for the infinite. Here shows itself the difference between system and schema. A schema gives only finite rubrics, but a systematic development proceeds from the inner infinity of the matter.

¹⁸Anf. Skr. S. 88–90. 1) $\cos x = 1 + A_1x + A_2x^2 + A_3x^3 + A_4x^4 \dots$ ($\cos 0 = 1$, thus: $A_0 = 1$) $\cos(-x) = 1 - A_1x + A_2x^2 - A_3x^3 + A_4x^4 \dots$ ($-x$ gives, in odd powers, minus, in even, plus). But $\cos(-x) = \cos x$; if the 2 series are to be equal, $+A_1 = -A_1$, $+A_3 = -A_3 \dots$, then the coefficients with odd indices must all be 0. If, in the presentation of the series for $\sin x$, one uses B instead of A , thus $\sin x = B_0 + B_1x + B_2x^2 + B_3x^3 \dots$, one obtains, since $\sin 0 = 0$, and B_0 consequently = 0, $\sin x = B_1x + B_2x^2 + B_3x^3 \dots$, (α), and $\sin(-x) = -B_1x + B_2x^2 - B_3x^3 + B_4x^4 \dots$ (β); but $\sin(-x) = -\sin x$ (Ramus. Trigonometri S. 8.), and according to (α) $= -B_1x - B_2x^2 - B_3x^3 \dots$. In order that the series (α) for $-\sin x$ may agree with (β) for $\sin(-x)$, $B_2, B_4 \dots$ must be 0. 2) From the cited series one procures oneself α) a development for $\cos x \cdot \cos z$, in which there occurs a term $A_m A_n x^m z^n$, β) a development for $\sin x \sin z$, in which there occurs a term $B_p B_q x^p z^q$, and γ) from trigonometry (Ramus. Trig. S. 9.) the two equations: $\cos x \cdot \cos z = \frac{1}{2} (\cos(x+z) + \cos(x-z))$ and $\sin x \cdot \sin z = \frac{1}{2} (\cos(x-z) - \cos(x+z))$, and then obtains from the series for cosine: $A_m A_n = A_{m+n} \frac{(m+n)(m+n-1) \dots (m+1)}{1 \cdot 2 \dots n}$, and from the series for sin, compared with the equations for cos, $B_p B_q = -A_{p+q} \frac{(p+q)(p+q-1) \dots (p+1)}{1 \cdot 2 \dots q}$. 3) The relations given lead to definite expressions for the coefficients, namely in the series for cosine: $A_m = \frac{(-1)^{\frac{m}{2}}}{1 \cdot 2 \dots m}$, $A_2 = -\frac{1}{2}$; in the series for sine: $B_p = \frac{(-1)^{\frac{p-1}{2}}}{1 \cdot 2 \dots p}$, $B_1 = +1$. In the cosine series one then has $A_2 = -\frac{1}{2}$, $A_4 = \frac{(-1)^{\frac{4}{2}}}{1 \cdot 2 \cdot 3 \cdot 4} = \frac{+1}{1 \cdot 2 \cdot 3 \cdot 4}$, $A_6 = \frac{(-1)^{\frac{6}{2}}}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} = \frac{-1}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6}$; in the sine series: $B_1 = 1$, $B_3 = \frac{(-1)^{\frac{3-1}{2}}}{1 \cdot 2 \cdot 3} = \frac{-1}{1 \cdot 2 \cdot 3}$, $B_5 = \frac{(-1)^{\frac{5-1}{2}}}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} = \frac{+1}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} \dots$

b) The Reciprocal Relation of the Functions.

In a schema nominal definitions hold, but motivated conceptual transitions in a system.

As an exact science, mathematics gives not merely exact definitions, but shows besides how the concepts form themselves, and how they operate. It is not the definition of a logarithm that teaches us what a logarithm is; we learn it provisionally, in that we see the connection in which the concept arises; we learn it thoroughly when we see the logarithmic function in its manifold transitions, in the consequences of its all-sided development.

In a schema mechanical division prevails, but in a system organic articulation.

As an exact science, mathematics maintains exact divisions; one may think of the division of the elementary functions into simple and composite, algebraic and transcendent, and the like. But the functions thus divided are not held fast within the finite limits of measured-out rubrics. One sees from the operations that the limit is transgressed, and yet precisely maintained, in that it is transgressed.

This holds, for example, of the limit between transcendent and algebraic equations. $a^x = b^y$ is transcendent; but through $x \log a = y \log b$ there takes place a transition to $y = \frac{x \log a}{\log b}$, which is algebraic.

The same holds of the limit between the exponential function and the logarithmic. In the series for the exponential function the coefficient A_1 was (II, 2) determined by a logarithmic expression, namely $A_1 = \frac{\log a}{\log e}$; one obtains, then:

$$a^x = 1 + \frac{\log a}{\log e} x + \left(\frac{\log a}{\log e} \right)^2 \frac{x^2}{1 \cdot 2} + \left(\frac{\log a}{\log e} \right)^3 \frac{x^3}{1 \cdot 2 \cdot 3} + \dots$$

Since now the logarithm of the base is always 1, one chooses the system whose base is $e = 2,718281828459 \dots$, and, denoting the logarithm in this system by l , there results

$$a^x = 1 + (l. a) x + (l. a)^2 \frac{x^2}{1 \cdot 2} + (l. a)^3 \frac{x^3}{1 \cdot 2 \cdot 3} \dots \dots$$

One thus not only passes, within the limit of the logarithm-category, over from the one system into the other; but one applies the logarithm-category in order to make

clear to oneself the power-category: the value of an exponential function is asked for, and answered with the value of a logarithmic one; the value of the logarithmic function is asked for, and answered with the value of the exponential.¹⁹

The same holds of the limit between the exponential function and the trigonometric ones.

If in the equation $a^x = 1 + (l \cdot a)x + (l \cdot a)^2 \frac{x^2}{1 \cdot 2} + (l \cdot a)^3 \frac{x^3}{1 \cdot 2 \cdot 3} + \dots$ one sets e instead of a , and recalls that $l \cdot e = 1$, one obtains:

$$e^x = 1 + \frac{x}{1} + \frac{x^2}{1 \cdot 2} + \frac{x^3}{1 \cdot 2 \cdot 3} + \dots$$

If in this equation one sets $x\sqrt{-1}$ instead of x , one obtains:

$$\begin{aligned} e^{x\sqrt{-1}} &= 1 + \frac{x\sqrt{-1}}{1} + \frac{(x\sqrt{-1})^2}{1 \cdot 2} + \frac{(x\sqrt{-1})^3}{1 \cdot 2 \cdot 3} + \frac{(x\sqrt{-1})^4}{1 \cdot 2 \cdot 3 \cdot 4} \dots \\ &= 1 + x\sqrt{-1} - \frac{x^2}{1 \cdot 2} - \frac{x^3\sqrt{-1}}{1 \cdot 2 \cdot 3} + \frac{x^4}{1 \cdot 2 \cdot 3 \cdot 4} \dots \end{aligned}$$

for the powers of $\sqrt{-1}$ form a period of 4 terms: $\sqrt{-1}$, -1 , $-\sqrt{-1}$, and $+1$, which repeat themselves to infinity.

If one now multiplies the equation $\sin x = x - \frac{x^3}{1 \cdot 2 \cdot 3} + \frac{x^5}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} - \frac{x^7}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} + \dots$ by $\sqrt{-1}$, one obtains $\sin x \sqrt{-1} = x\sqrt{-1} - \frac{x^3\sqrt{-1}}{1 \cdot 2 \cdot 3} \dots$, which value corresponds precisely to the 2nd and 4th terms, and so forth, in the series for $e^{x\sqrt{-1}}$, while the value for $\cos x$, namely $1 - \frac{x^2}{1 \cdot 2} + \frac{x^4}{1 \cdot 2 \cdot 3 \cdot 4} \dots$, corresponds to the 1st, 3rd, and 5th terms, and so forth.

Thus, in $e^{x\sqrt{-1}} = \cos x + \sqrt{-1} \sin x$, and, when one sets $-x\sqrt{-1}$ instead of $x\sqrt{-1}$, in $e^{-x\sqrt{-1}} = \cos x - \sqrt{-1} \sin x$, the exponential function is expressed by the trigonometric; from which it further follows that the trigonometric can likewise be expressed by the logarithmic. For if $e^{x\sqrt{-1}} = \cos x + \sqrt{-1} \sin x$, then also $x\sqrt{-1} = l \cdot (\cos x + \sqrt{-1} \sin x)^{20}$.

The systematic intercourse between the exponential function and the trigonometric

¹⁹The equation cited under a): $\log(1+x) = \log e \left(x - \frac{x^2}{1 \cdot 2} + \frac{x^3}{1 \cdot 2 \cdot 3} - \frac{x^4}{1 \cdot 2 \cdot 3 \cdot 4} \dots \right)$ is developed thus: 1) $(1+x)^y = 1 + \frac{y}{1}x + \frac{y(y-1)}{1 \cdot 2}x^2 + \frac{y(y-1)(y-2)}{1 \cdot 2 \cdot 3}x^3 \dots$ 2) $(1+x)^y = 1 + \frac{\log(1+x)}{\log e}y + \log \left(\frac{1+x}{\log e} \right)^2 \frac{y^2}{1 \cdot 2} \dots$ by ordering these series according to increasing powers of y , and then comparing them.

²⁰"The trigonometric lines are, as functions of the circular arc, reducible to the exponential function, and the circular arc, as a function of the trigonometric lines, reducible to the logarithm." Ramus. *Algr. Functsl.* S. 15.

ones is thus brought about by the application of an imaginary magnitude, $\sqrt{-1}$; but an imaginary magnitude belongs, according to the definition, to the impossible. By the in-itself impossible, then, a real reduction has become possible: how is this now to be understood?

If anyone should answer: $\sqrt{-1}$ is both impossible and yet possible; then formal logic would perhaps interpret the answer thus: $\sqrt{-1}$ is in one respect impossible, in so far, namely, as there is here required a number which, raised to the 2nd power, should give -1 , in another respect, on the contrary, possible, in that one, by, for example, raising $\sqrt{-1}$ to the 2nd power, obtains a real magnitude, -1 . This might then be an example of the maxim: *diversus respectus tollit omnem contradictionem*.

But with all respect for *diversus respectus*, such a resolution of the contradiction is nevertheless only formal, and the merely formal is the superficial. With this superficiality science cannot set itself at rest: mathematics proceeds to operations; philosophy follows after, and looks at the dialectical.

If one could, with the logical maxim *possibile est, quod non implicat contradictionem*, as regards the concepts, keep to nominal definitions, one would soon be done with the separation of the possible and the impossible²¹; but a thorough separation is conditioned by systematics.

Mathematics teaches us, when we follow its systematic development, that much is possible which at first glance seems impossible, and, conversely, much at bottom impossible which an uninitiated person easily takes for possible.

Yes, so dialectical is the limit of possibility, even in the mathematical domain, that the exact science itself in many points is still scarcely able to hold it fast.²²

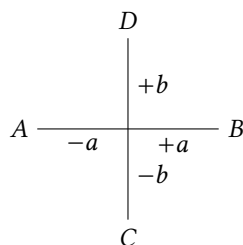
That the limits are maintained in that they are transgressed shows, moreover, how definitions and transitions, mechanical division and infinite articulation, condition one another in a scientific system. In so far as the several functions are placed as if in

²¹In this way a round quadrangle would indeed be impossible, but hardly a regular heptahedron.

²²In order not to dwell on questions such as: is it possible to find a comprehensive law for the determination of the prime numbers? is it possible to find a finite solution for the general equation of the n th degree? is it absolutely impossible to bring the quadrature of the circle under a finite expression, absolutely impossible that π can be periodic? and so forth — we will in this connection bring out a little more closely the dispute concerning the imaginary magnitudes. Thus: 1) is $\sqrt{-1}$ an absolute impossibility? Since $(-a)$ is produced from $(+a)$ by multiplying this magnitude by (-1) , one can make clear to oneself the meaning of this multiplication by rotations.

rubrics side by side, and held against one another, they are placed realistically: the more firmly the finite opposition is held, the stronger is the realism. Mathematical practice, however, itself subdues this realism, in that it lets each of the separated functions take up the others into its self-development. Each function thus returns, through the others, into itself, has thus the others as moments in its determination of concept and its content; this is the idealistic. For the realist the relations are finite, for the idealist they are infinite: in a carried-through systematics the finite relations become grounded in the infinite.

c) Taylor's Series.



A rotation of 90° is then denoted by $+a\sqrt{-1}$; of 180° by $+a\sqrt{-1}\sqrt{-1} = a(-1) = -a$; of 270° by $a(-1)\sqrt{-1} = -a\sqrt{-1}$; of 360° by $-a\sqrt{-1}\sqrt{-1} = -a(-1) = +a$. But then one must, to be sure, combine such heterogeneous concepts as multiplication and rotation. If one has, the origin of the coordinate system being at the centre, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ as the equation of the ellipse, then one can, by setting $b = \sqrt{-1} b'$, and consequently $b^2 = -b'^2$, obtain: $\frac{x^2}{a^2} - \frac{y'^2}{b'^2} = 1$, which is the equation of the hyperbola. The impossible and therefore imaginary $\sqrt{-1}$ thus becomes, after all, in a way real, and therefore possible. 2) Are logarithms of negative magnitudes possible? On a fleeting consideration it should seem as if there were nothing in the way. One has $\log(a^2) = 2 \log a$, and $\log(-a)^2 = \log(a^2)$; how near, then, lies the conclusion that $\log(-a)^2$ must also be $= 2 \log(-a)$, and yet the conclusion is false. One has, when a and b are positive, $\sqrt{a} \cdot \sqrt{b} = \sqrt{a \cdot b}$; but since $\sqrt{-a}$ is impossible, the above equation does not hold for negative magnitudes. One cannot say $(\sqrt{-a})^2 = \sqrt{-a} \sqrt{-a} = \sqrt{(-a)^2} = \sqrt{a^2} = \pm a$, for it is only $\sqrt{-a}$ which, raised to the 2nd power, gives $-a$. In the dispute against Bernoulli, Leibnitz saw the truth, but lacked the means to disarm his opponent; in the dispute against d'Alembert, Euler applied the formulas: $a^x = 1 + (l.a)\frac{x}{1} + (l.a)^2\frac{x^2}{1 \cdot 2} + \dots$, $e^{x\sqrt{-1}} = \cos x + \sin x \sqrt{-1}$, and $l(\cos x + \sin x \sqrt{-1}) = x \sqrt{-1}$, and found in these formulas points of departure for clear proofs. For if one sets the formula: $e^{(2p\pi+y)\sqrt{-1}} = \cos(2p\pi+y) + \sin(2p\pi+y)\sqrt{-1}$, p = any whole number whatever; and sets further $a = r \cos(2p\pi+y)$, $b = r \sin(2p\pi+y)$, one obtains, by the roundabout way of the reductions (jvfr. Ramus. Algr. Functsl. S. 92-94), the following formulas for comparison: (I) $l.a^2 = l^1 a^2 + 2p\pi \sqrt{-1}$; (II) $l.a^2 = 2l^1 a + 2p\pi \sqrt{-1}$; (III) $2l.a = 2l^1 a + 4p^1 \pi \sqrt{-1}$ (in that the equation $l.a = l^1 a + 2p^1 \pi \sqrt{-1}$ is multiplied by 2); (IV) $2l(-a) = 2l^1(a) + 2(2p^{11}+1)\pi \sqrt{-1}$ (since one has $l(-a) = l^1 a + (2p^{11}+1)\pi \sqrt{-1}$). If IV is compared with III, then $2l(-a) = 2l(a)$ when $4p^1 = 2(2p^{11}+1)$, or when $2p^1 = 2p^{11}+1$; but this is impossible, since an even number can never become equal to an odd one. According to the other comparisons one finds that the terms which $l(-a)$ has in common with $l.a$ in the infinite series are altogether imaginary. A real logarithm of $-a$ is thus, after all, an impossibility.

The systematic connection of finite terms, which articulates the inner infinity, is in all points immediate and yet infinitely mediated.

If, however, anyone should say: “our cognition is thus immediate only because it is so infinitely mediate”, such an utterance would easily be taken as a proof of subtlety and dialectical wordplay.

Yet the validity of the proposition set up will precisely become immediately evident, once the understanding has first been mediated.

In a chain of material links it holds, to be sure, that while the first and second links are in immediate connection, the first and third links are only in a mediate, the first and fourth in a still more mediate connection. But with such a parcelling-out of the immediate and the mediate the concept “inner infinity” is incompatible.

In the intellectual connection one speaks, to be sure, also of the connection of the extreme terms by middle terms; but the deeper the insight goes, the more the finite middle terms, as determinations of concept, are dissolved in the intimate unity of the extreme terms.

In order not to be dazzled by the mediacy in the entering middle terms, one must look at the light-effect, and make clear to oneself the difference between the transparent and the non-transparent.

As regards the sensible light-effect, it is the obscurity of the middle terms, the finitude of the means, that lets us feel the mediate.

He who would know figures that move behind a partition-wall must by a round-about way procure himself information, and thereby marks the mediate. If now it is not a wall, but a translucent pane, that separates the beholder from the objects, the mediate already begins to disappear: he sees the shadows move immediately, although the pane is perceived as a distancing middle term. If the pane is of purely transparent glass, the middle term does not trouble at all; and if the eye is armed besides with the glass of the telescope, then we have here an example of middle terms which, instead of removing the corresponding extreme terms, the beholder and the object, on the contrary bring them nearer to one another, and, instead of binding them by the mediate, even let them meet in heightened immediacy.

If the presupposition in question is evident, then its nearest consequence is also evi-

dent. Once one has understood what is meant by $A_m A_n = A_{m+n} \frac{(m+n)(m+n-1)\dots(m+1)}{1 \cdot 2 \cdot 3 \dots n}$, then it is also immediately evident that $A_m A_1$ must be, for $n = 1$, $= A_{m+1} \frac{m+1}{1}$; if the equation $A_m A_1 = A_{m+1} \frac{m+1}{1}$ is immediately evident, then the equation $A_{m+1} = \frac{A_m A_1}{m+1}$ is also immediately evident.

But thus it stands with all transitions in the higher as in the lower mathematics. Each proposition must, in its peculiar connection, be immediately evident, and yet in this peculiar connection there is precisely the mediate. The more the connection is cleared up, the more transparent the middle terms become, the more, therefore, will the mediate also merge into higher immediacy.

For the mathematically educated it is just as easy to run through the proof of $A_m A_n = A_{m+n} \frac{(m+n)(m+n-1)\dots(m+1)}{1 \cdot 2 \cdot 3 \dots n}$ in the above-cited connection, as for the beginner to understand the inference: $A > B$, $B > C$, thus $A > C$.

If we consider the equation:

$$f(x+h) = A_0 + A_1(x+h) + A_2(x+h)^2 + A_3(x+h)^3 \dots$$

as an infinite chain of finite terms, then it can be proved that in such a chain there is the possibility of an infinite mediation of the connection of the terms. The method, the way of mediation, is at once a roundabout way and a short cut.

Since $f(x+h)$ is not yet more closely determined, one can, in the first place, set the condition that the otherwise undetermined coefficients $A_0, A_1, A_2 \dots$ shall be functions, namely different functions of x , not of h , and, to recall the fulfilment of the condition, write the series thus:

$$f(x+h) = f(x) + f_1(x)h + f_2(x)h^2 + f_3(x)h^3 + \dots f_m(x)h^m + \dots = X_0 + X_1h + X_2h^2 + X_3h^3 \dots + X_mh^m \dots$$

In order to secure for ourselves an all-sided systematic connection, we set, in the second place, the condition that the infinite series be developed in such a way that its terms can again be developed in infinite series.

In order that this condition may be fulfilled, we introduce, in the third place, besides h still an increment k , and thereby procure for ourselves a double development, in

that k is on the one side added to h , on the other side to x .

Finally, in the fourth place, the condition must still be added, that the function be so chosen that one expressly obtains: $f(x + \overline{h + k}) = f(\overline{x + k} + h)$.

These, then, are our nearest presuppositions; we shall now test the consequences of these presuppositions.

In the series (I):

$$f(x + \overline{h + k}) = X_0 + X_1(h + k) + X_2(h + k)^2 + \dots X_{m+n}(h + k)^{m+n} \dots$$

there can, from the last-cited term, be developed an infinite series which, among its countless terms, must contain the following:

$$X_{m+n} \frac{(m + n)(m + n - 1) \dots (m + 1)}{1 \cdot 2 \dots n}$$

In the series (II), with its series:

$$\begin{aligned} f(\overline{x + k} + h) &= f(x + k) (= X_0 + X_{0,1}k + X_{0,2}k^2 \dots X_{0,n}k^n \dots) \\ &+ f_1(x + k)h (= X_1h + X_{1,1}hk + \dots X_{1,n}hk^n \dots) \\ &+ f_2(x + k)h^2 (= X_2h^2 + X_{2,1}h^2k \dots X_{2,n}h^2k^n \dots) \\ &\vdots \\ &+ f_m(x + k)h^m (= X_mh^m + X_{m,1}h^mk + \dots X_{m,n}h^mk^n \dots) \end{aligned}$$

one has, then, $X_{m,n}h^mk^n$, according to the method of undetermined coefficients,

$$= X_{m+n} \frac{(m + n)(m + n - 1) \dots (m + 1)}{1 \cdot 2 \dots n}.$$

In such combinations the mediate prevails: the way of the method is a roundabout way.

That now this roundabout way is precisely a short cut, in that the extreme terms, distanced by middle terms, are brought infinitely near to one another, can in this connection just as easily be demonstrated:

One has, according to (I), $X_1(h + k) = X_1h + X_1k$, and, according to (II), $X_{0,1}k$, and

thus, according to the method of undetermined coefficients, $X_{0,1} = X_1$; from this it is seen that $X_{m,1}$ arises from X_m in the same way as $X_{0,1} = X_1$ from X_0 .

If one now sets $X_0 = f(x)$, and $X_1 = \frac{f'(x)}{1}$, then one obtains, for $m = 1, 2, \dots = n, \dots$

$$\begin{aligned} X_{1+1} = X_2 &= \frac{X_{1,1}}{2} = \frac{\frac{f''(x)}{1}}{2} = \frac{f''(x)}{1 \cdot 2} \\ X_{2+1} = X_3 &= \frac{X_{2,1}}{3} = \frac{\frac{f'''(x)}{1 \cdot 2}}{3} = \frac{f'''(x)}{1 \cdot 2 \cdot 3} \\ X_n &= \frac{X_{n-1,1}}{n} = \frac{f^n(x)}{1 \cdot 2 \dots n} \end{aligned}$$

If, then, one knows the manner in which X_1 arises from X_0 , one knows at the same time the manner in which each succeeding term arises from its preceding one. If the relation between the first and second terms is evident, then at the same time the relation between the first and third, first and fourth, first and thousandth terms becomes evident. The short cut is conditioned by this, that the mediacy is raised to higher immediacy — in the depth of insight.

If now the coefficients are, on the conditions given, determined as functions of x , the series:

$$f(x+h) = f(x) + f_1(x)h + f_2(x)h^2 + f_3(x)h^3 + \dots f_m(x)h^m \dots$$

can be written:

$$f(x+h) = f(x) + f'(x)\frac{h}{1} + f''(x)\frac{h^2}{1 \cdot 2} + f'''(x)\frac{h^3}{1 \cdot 2 \cdot 3} + \dots f^m(x)\frac{h^m}{1 \cdot 2 \cdot 3 \dots m} \dots$$

This so-called Taylorian series can, when one, after having produced $f'(x)$, $f''(x)$... from $f(x)$, sets $x = 0$, and $h = x$, be transformed again into Maclaurin's series:

$$f(x) = f(0) + f'(0)\frac{x}{1} + f''(0)\frac{x^2}{1 \cdot 2} + \dots f^m(0)\frac{x^m}{1 \cdot 2 \cdot 3 \dots m} \dots$$

If, then, we hold fast to the identity of roundabout way and short cut, then a retrospect now also shows that the infinite mediation of the method is due at once to the necessary

movement of the matter and the free movement of thought.

We have the formula: $a^x = 1 + \frac{A_1 x}{1} + \frac{A_1^2 x^2}{1 \cdot 2} + \frac{A_1^3 x^3}{1 \cdot 2 \cdot 3} \dots$; it is expressly our task to find A_1 ; but the definite expression for A_1 by no means follows from the formula in the same way as from $A > B, B > C$ follows $A > C$.

The solution of the task is necessarily conditioned by the given connection, but it is nevertheless thought that sets itself the task, and finds out the way.

To set $x = \frac{1}{A_1}$ looks at first glance like an ingenious device, for it is an expedient that freedom has found; but, more closely considered, the expedient is prescribed. Freedom's choice is grounded in the necessity of the matter.

We have the formulas: $a^x = 1 + (l.a)x + (l.a)^2 \frac{x^2}{1 \cdot 2} + (l.a)^3 \frac{x^3}{1 \cdot 2 \cdot 3} \dots$ and $e^x = 1 + \frac{x}{1} + \frac{x^2}{1 \cdot 2} + \frac{x^3}{1 \cdot 2 \cdot 3} \dots$. The task is to find a systematic connection between the exponential function and the trigonometric ones. The necessary middle term lies in the matter; but it takes a Euler to discover the middle term.

We have in the formula:

$$f(x+h) = A_0 + A_1(x+h) + A_2(x+h)^2 + A_3(x+h)^3 \dots$$

the possibility of countless series-expansions. The conditions that are set, the directions that are followed, the whole method, are here just as little an immediate consequence of necessity as a chance product of arbitrariness.

It is the matter that yields the conditions, but free thought that chooses the conditions; it is the matter that furnishes the presuppositions, but free thought that orders the presuppositions; it is the matter that supplies the means, but free thought that fixes the goal. The goal is, in this connection, an analysis of the infinite.

III.

The Analysis of the Infinite.

In the approach toward an Unattainable the problem of infinity first found its contemplable expression; the inner infinity of the finite relations then showed us the way to the solution of the problem.

That such a solution concerns not only mathematics, but in the highest degree philosophy, can already be seen from the contradiction in which the knot is tied. The contradiction is that the infinite is at an unattainable remoteness (the asymptote), because it is everywhere so infinitely near to the finite (the trigonometric functions).

If we follow, as hitherto, the mathematical track, the incessant approach is to be more closely determined by the function-limit, the *Limit*²³, the limit toward which a function of x converges, as x either increases or decreases to infinity. As regards the, so to speak, omnipresence of the infinite in the finite, we are then referred to the *Differential*, the infinitude-difference that arises between $f(x)$ and $f(x + h)$, when h is regarded as an infinitely small increment to x ; finally to the *Integral*, the expression which shows how one, through the roundabout ways of the infinite, finds the short cut back to the finite.

1) The Function-Limit.

In the equation $\text{Lim } f(x) = f(\infty) = c$, c is the value, the limit, to which $f(x)$ more and more approaches, as x becomes greater and greater — a limit which, however, is first attained when x becomes $= \infty$, thus by infinite approach.

Example:

$$f(x) = \frac{a + a_1x + a_2x^2 + \dots a_nx^n}{b + b_1x + b_2x^2 + \dots b_nx^n}$$

So long as x is only a finite magnitude, none of the terms can be thrown away; however, one can, by dividing by x^n , transform the whole quotient without altering its value.

Thus:

$$f(x) = \frac{\frac{a}{x^n} + \frac{a_1}{x^{n-1}} + \frac{a_2}{x^{n-2}} + \dots a_n}{\frac{b}{x^n} + \frac{b_1}{x^{n-1}} + \frac{b_2}{x^{n-2}} + \dots b_n}$$

The transformation shows that those terms in numerator and denominator which are divided by x acquire less and less value, the more x grows. If now one sets $x = \infty$, one

²³Ramus. Algbr. Funct. S. 78–79.

obtains:

$$f(\infty) = \frac{\frac{a}{\infty^n} + \frac{a_1}{\infty^{n-1}} + \frac{a_2}{\infty^{n-2}} + \dots a_n}{\frac{b}{\infty^n} + \frac{b_1}{\infty^{n-1}} + \frac{b_2}{\infty^{n-2}} + \dots b_n} = \frac{0 + 0 + 0 + \dots a_n}{0 + 0 + 0 + \dots b_n} = \frac{a_n}{b_n} = c$$

In the equation:

$$\lim f(x) = f(0) = c'$$

c' denotes the value, the limit, to which the function approaches, as x becomes smaller, but which is only reached in infinite approach, in so far as it is first reached when $x = 0$.

Example. From $f(x) = \frac{a + a_1x + a_2x^2 + \dots a_nx^n}{b + b_1x + b_2x^2 + \dots b_nx^n}$ it is seen that all the terms multiplied by x must acquire a smaller value, the smaller x becomes. If x at last becomes $= 0$, one gets:

$$f(0) = \frac{a + a_1 \cdot 0 + a_2 \cdot 0^2 + \dots a_n \cdot 0^n}{b + b_1 \cdot 0 + b_2 \cdot 0^2 + \dots b_n \cdot 0^n} = \frac{a}{b} = c'.$$

But, one might object, one is so far from yet seeing any difference between ∞^n and ∞^1 , between 0^n and 0^1 , that it even seems as if 0 and ∞ , even when they are introduced by the way of approach, nevertheless continue to obliterate all difference between the finite, definite magnitudes.

In order to grasp the relativity which the approach introduces into the infinite, we must consider the infinite magnitudes 0 and ∞ , partly in relation to the finite, partly in relation to one another.

$$a > b; \quad \text{but } a \cdot \infty = b \cdot \infty, \quad a \cdot 0 = b \cdot 0.$$

$$\text{Lim}(x) = \infty; \quad \text{Lim} \left(\frac{1}{x} \right) = \text{Lim} \frac{1000}{x} = 0.$$

$$\lim(x) = 0; \quad \lim \frac{1}{x} = \lim \frac{1000}{x} = \infty.$$

Thus every finite magnitude will, in comparison with the infinitely great, be vanishing; and likewise the infinitely small will be vanishing in comparison with every finite

magnitude. In so far as the finitude-limit has vanished, 0 and ∞ still maintain their rank with the infinite; but in so far as the vanished points back to a vanishing, and this vanishing to the approach, 0 and ∞ descend from the absolute height of infinity, and enter into calculation with finite magnitudes.

We can make the matter clear thus. In the formula $\frac{x}{x} = 1$, x may well change, become greater and greater, smaller and smaller, without the quotient in question changing. $\lim_{x \rightarrow \infty} \frac{x}{x} = \frac{\infty}{\infty}$ is thus no longer an indeterminate expression, but quite determinately = 1; equally determinate is $\lim_{x \rightarrow 0} \frac{x}{x} = \frac{0}{0} = 1$. What holds of the quotient $\frac{x}{x'}$, holds also of $\frac{x^2}{x}$, $\frac{x^3}{x}$, $\frac{x^n}{x}$, and so forth. Thus: since $\frac{x^2}{x} = x$, $\lim_{x \rightarrow \infty} \frac{x^2}{x} = \frac{\infty^2}{\infty} = \infty$, and $\lim_{x \rightarrow 0} \frac{x^2}{x} = \frac{0^2}{0} = 0$. Since $\frac{x^n}{x} = x^{n-1}$, then $\lim_{x \rightarrow \infty} \frac{x^n}{x} = \frac{\infty^n}{\infty} = \infty^{n-1}$, and $\lim_{x \rightarrow 0} \frac{x^n}{x} = \frac{0^n}{0} = 0^{n-1}$.

In order to see how the contradiction which lies in the concept of infinite magnitude is resolved in the mathematical presentation, we can for a moment let the infinite approach halt at a finite limit. Where this limit is set is, as regards the concept, indifferent, since it is set only in order to be immediately transgressed.

The magnitude 1,000,000 represents for a moment ∞ ; the quotient $\frac{1}{1,000,000}$ will then necessarily represent 0.

In accordance with the above formulas we shall thus obtain:

$$\begin{aligned} \frac{1,000,000}{1,000,000} &= \frac{\infty}{\infty} = 1; & \frac{\frac{1}{1,000,000}}{\frac{1}{1,000,000}} &= \frac{0}{0} = 1. \\ \frac{(1,000,000)^2}{1,000,000} &= \frac{\infty^2}{\infty} = 1,000,000 = \infty; \\ \frac{\left(\frac{1}{1,000,000}\right)^2}{\frac{1}{1,000,000}} &= \frac{0^2}{0} = \frac{1}{1,000,000} = 0. \\ \frac{(1,000,000)^3}{1,000,000} &= \frac{\infty^3}{\infty} = 1,000,000^2 = \infty^2; \\ \frac{\left(\frac{1}{1,000,000}\right)^3}{\frac{1}{1,000,000}} &= \frac{0^3}{0} = \left(\frac{1}{1,000,000}\right)^2 = 0^2 \end{aligned}$$

The objection that 1,000,000 is infinitely far from ∞ , and $\frac{1}{1,000,000}$ consequently equally far from 0, means nothing; since the finitude-limit is set only in order to be transgressed, one will not only be able to let $1,000,000^{1,000,000}$ instead of 1,000,000 represent ∞ , but — what is here decisive — also see that the terms of the relation can be changed to

infinity, without the definite relation being thereby in the least altered.

So far are 0 and ∞ from obliterating all difference of magnitude, that they conversely give the finite difference an infinite significance. If there is here between x^2 and x a definite finite difference, because x is finite, then there is now also between ∞^2 and ∞ a definite infinitude-difference, and likewise between 0^2 and 0.

Instead of bursting the finite, and, with the annihilating superior power of the absolute, so to speak, crushing the relative, 0 and ∞ are rather, to use a figurative expression, like good gods descended into finitude, to exalt the relative and to preserve the invariable in all changes.

The infinite becomes finite, that the finite may become infinite. 0 and ∞ are no longer absolute; here there is no more any absolute difference between 0, 1, ∞ . In comparison with 0, $1 = \infty$; compared with ∞ , $1 = 0$. In relation to ∞^2 , $\infty = 0$; in relation to 0^2 , $0 = \infty$.

The same law which maintains a definite relation of finite terms continues still, after the finitude of the terms has vanished, be it in 0 or in ∞ , to maintain the same relation; but when it is demonstrable that the relation for the infinite becomes the same as for the finite, the unattainable can be said to be attained.

The dialectic that the infinite is as the finite, and the unattainable in every Now attainable, mathematics satisfies in manifold ways by its operations with the infinite series of the formula $f(x + h)$.

If one has, to choose a definite example, a function of one variable $f(x) = \frac{\varphi(x)}{\psi(x)}$, which for a special value $x = a$ assumes the form $\frac{0}{0}$, since $\varphi(a) = 0$ and $\psi(a) = 0$; then the true value of such an expression can be determined by seeking the developments of $\varphi(a + h)$ and $\psi(a + h)$ according to increasing powers of h ,

$$\varphi(a + h) = A_1 h^{\alpha_1} + A_2 h^{\alpha_2} + A_3 h^{\alpha_3} + \dots$$

$$\psi(a + h) = B_1 h^{\beta_1} + B_2 h^{\beta_2} + B_3 h^{\beta_3} + \dots$$

where α and β are positive, since $h = 0$ is to reduce both developments to 0. One then has:

$$f(a + h) = \frac{\varphi(a + h)}{\psi(a + h)} = h^{\alpha_1 - \beta_1} \frac{A_1 + A_2 h^{\alpha_2 - \alpha_1} + A_3 h^{\alpha_3 - \alpha_1} + \dots}{B_1 + B_2 h^{\beta_2 - \beta_1} + B_3 h^{\beta_3 - \beta_1} + \dots}$$

By setting $h = 0$, one obtains²⁴:

$$f(a) = \frac{\varphi(a)}{\psi(a)} = 0^{\alpha_1 - \beta_1} \frac{A_1}{B_1}$$

Three cases are now possible: I) $\alpha_1 > \beta_1$; $\alpha_1 - \beta_1 = +n^1$. II) $\alpha_1 < \beta_1$; $\alpha_1 - \beta_1 = -n^1$. III) $\alpha_1 = \beta_1$; $\alpha_1 - \beta_1 = 0$. Thus:

$$\begin{aligned} \text{I) } f(a) &= 0^{n^1} \cdot \frac{A_1}{B_1} = \frac{0^{n^1} A_1}{B_1} = \frac{0}{B_1} = 0. \\ \text{II) } f(a) &= 0^{-n^1} \cdot \frac{A_1}{B_1} = \frac{1}{0^{n^1}} \cdot \frac{A_1}{B_1} = \frac{A_1}{0} = \infty. \\ \text{III) } f(a) &= 0^0 \cdot \frac{A_1}{B_1} = 1 \cdot \frac{A_1}{B_1} = \frac{A_1}{B_1}. \end{aligned}$$

From the magnitude h it is seen here what it means that the finite can be as 0, and 0 as a finite.

By infinite approach one sets $h^{\alpha_1 - \beta_1} = 0^{\alpha_1 - \beta_1}$; for if 0 were an absolute negation, the expression $0^{\alpha_1 - \beta_1}$ would be entirely meaningless. In so far as here there is only an approach to 0, not the absolute 0, the infinite is still, as before, the unattainable-by-approach.

But the infinite approach cannot, on the other hand, be understood in such a way that h retains a finite, though only quite insignificant, value, and is therefore, with a certain approximateness and as it were at random, set = 0. Here, in the equation $h^{\alpha_1 - \beta_1} = 0^{\alpha_1 - \beta_1}$, there is not the faintest tinge of the merely approximate, not the least trace of the inexact. The equation $h^{\alpha_1 - \beta_1} = 0^{\alpha_1 - \beta_1}$ is, in the concept, exact, like the equation $A = A$.

But if the equation is exact, then the infinite has, in the function-limit, gone together with the finite, and the unattainable is thus after all attained.

2) The Differential.

In so far as here there is talk of approach, the infinite lies constantly outside the finite, and is therefore the unattainable. For the attainment of the unattainable a leap

²⁴Ramus. Differential- og Integralregning 1844. S. 12–13.

must here, then, from this standpoint, be made; but, in order that not too much be leaped over either, the leap must in turn be sufficiently prepared by approach.

This way of viewing has its significance in finite calculations. How many decimals one is, for example, to take of the number π depends on the accuracy with which the calculation is carried out; one can in certain cases be content with 5 decimals, one can also go to 10, to 50, to 100, and more. But in all such things approximateness prevails: a finite bargaining about leap and approach is therefore misleading when it is a question of insight into the infinite.

It is then also easy, in mathematics, to demonstrate how the infinite distinguishes itself from the finite only as if by a leap, in so far as it itself leaps out of the finite.

From $a = a$ there follows incontestably $a - a = 0$, and conversely from $a - a = 0$ there follows incontestably $a = a$. If we interpret these equations in logical judgments, the difference between the finite and the infinite is seen as a difference between the positive and the negative, between that which is represented and that which is thought.

Conceived as a positive judgment, $a = a$ declares that the finite is what it is, and relates to itself immediately, without any negation; conceived as a negative judgment, $a - a = 0$ declares that the finite, in order to be able to exclude the negation and relate to itself, must secure the relation by the negation, and thus after all take up the negation into the relation: a is a representation, 0 a concept.

In the quotient $\frac{1}{a-a} = \frac{1}{a} + \frac{1}{a} + \frac{1}{a} + \dots$ infinity is represented as a self-continuing finitude, an endless finitude, a finite that never comes to an end; in $\frac{1}{a-a} = \frac{1}{0} = \infty$, on the other hand, thought sees the infinite as a concept expressed in the quotient, an inner determination of relation, for which the finite terms are only moments.

In the equation $2 = 2$, a finite is represented without regard to the infinite; in the equation:

$$\frac{1}{1 - \frac{1}{2}} = 1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \dots$$

an infinite is represented by means of the finite; in the quotient:

$$\frac{1}{1 - \frac{1}{2}} = 1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \dots = 2$$

the identity of the finite and the infinite is to be seen as in a concept.

In order to exclude the approximate and hold fast the concept, mathematics has accordingly, with sharpened attention, brought out the mark of the convergence of infinite series²⁵.

The difference between divergent and convergent series is, together with the difference between infinite magnitudes of lower and higher order — 0^2 vanishing toward 0 , ∞ vanishing toward ∞^2 — grounded in the “function-limit”, and this determination of limit is therefore an indispensable presupposition for the determination of the differential.

That the differential must precisely be determined by a function-limit, in which the finite is united with the infinite as with 0 , that is, as with its own negation, can, according to the direction of mathematics, be presented thus: if $f(x) = y$, then $f(x + h) = y + k$; if, for example, $f(x) = x^2 = y$, then $f(x + h) = (x + h)^2 = x^2 + 2xh + h^2 = y + k$. Since now $x^2 = y$, then k does not become $= h$, but $= 2xh + h^2$.

If $h = 0$, there is here nothing at all of the finite, no increment for x ; then $f(x + 0) = f(x)$, then also $k = 0$; here there is no difference, thus no differential. If h is finite, then the difference between $f(x + h)$ and $f(x)$ is also finite; here there is then a difference, a finite magnitude k , but nothing of the infinite, no differential.

a) The Function-Limit and the Differential Coefficient.

As an infinitely small increment to x , h must be both something and nothing, dx , the differential of x ; as an infinitely small increment to y , k must likewise be both something and nothing, dy , the differential of y .

In relation to any finite magnitude whatever, dx and dy are both $= 0$; but to one another they are nevertheless in a quite definite relation: $\frac{dy}{dx} = \frac{0}{0}$, just as definite as, for example, $\frac{y}{x}$ or as $\frac{1}{2}$ or as $\frac{2}{3}$... Here shows itself the inner connection between

²⁵“In every convergent series” ($s_n = u_0 + u_1 + u_2 \dots + u_n$) “ $\text{Lim } u_n = 0$, but this is only a necessary, not a sufficient condition for convergence, for an infinite number of positive terms can give a sum $= \infty$. On the other hand, when r_n denotes the sum of all the terms following after u_n , u_{n+1} , u_{n+2} , u_{n+3} ... to infinity, which is called the remainder of the series, then $\text{Lim } r_n = 0$ will express both a necessary and a sufficient condition for the convergence of the series, for one can then always make n so great that r_n is smaller than any given magnitude whatever, whereby s_n is brought as near as one wills to a certain limit.” Ramus. *Algebr. F. S.* 79–80.

the function-limit and the differential quotient, or, as it is also called, the differential coefficient — a connection which is articulated in the application of Taylor's series.

From

$$f(x+h) = f(x) + f'(x) \frac{h}{1} + f''(x) \frac{h^2}{1 \cdot 2} + f'''(x) \frac{h^3}{1 \cdot 2 \cdot 3} + \dots$$

one obtains: $f(x+h) - f(x) = f'(x) \frac{h}{1} + f''(x) \frac{h^2}{1 \cdot 2} + f'''(x) \frac{h^3}{1 \cdot 2 \cdot 3} \dots$, and, by dividing by h ,

$$\frac{f(x+h) - f(x)}{h} = f'(x) + f''(x) \frac{h}{1 \cdot 2} + f'''(x) \frac{h^2}{1 \cdot 2 \cdot 3} + \dots$$

So long as h is a finite magnitude, no term of the infinite series can drop away without an inexactness showing itself; if, on the other hand, h is set = 0, all terms drop away with the exception of the first, $f'(x)$. Since now, as we have seen, h cannot immediately be set = 0, unless one will have $f(x+0) = f(x)$, and thereby annihilate dx ; there is set, in virtue of the law that holds both for the finite and the infinite,

$$\lim \frac{f(x+h) - f(x)}{h} = \lim \frac{y+k-y}{h} = \lim \frac{k}{h} = \frac{dy}{dx} = f'(x),$$

or the first differential coefficient of $f(x)$.

Conceived as a function-limit, $\lim f(h) = f(0)$, $f'(x)$ is, taken singly, to be seen under the aspect of an unattainable; conceived as a differential quotient, $f'(x)$ means that the unattainable is nevertheless after all attained, since the quotient $\frac{dy}{dx}$ is an exactly determined function.

If we look back at $f(x) = x^2 = y$, $f(x+h) = (x+h)^2 = y+k$, then, since $\frac{(x+h)^2 - x^2}{h} = 2x + h$,

$$\lim \frac{(x+h)^2 - x^2}{h} = \lim \frac{k}{h} = \frac{dy}{dx} = \frac{0}{0} = 2x.$$

b) Differentiation²⁶.

²⁶On the principles for the differentiation of explicit functions cf. Ramus. Algr. F. S. 99–108.

If $f'(x) = \frac{dy}{dx}$, and thus $dy = f'(x) dx$, then $f'(x+h) = \frac{dy}{dx} + k'$, thus:

$$\lim \frac{f'(x+h) - f'(x)}{h} = \lim \frac{\frac{dy}{dx} + k' - \frac{dy}{dx}}{h} = \lim \frac{k'}{h};$$

$= f''(x)$; but $k' = d \cdot \frac{dy}{dx}$, thus

$$\lim \frac{k'}{h} = \frac{d \cdot \frac{dy}{dx}}{dx},$$

which is written $\frac{d^2y}{dx^2}$, and $d^2y = f''(x) dx^2$; in the corresponding way the third differential coefficient is developed:

$$\frac{d \cdot \frac{d^2y}{dx^2}}{dx} = \frac{d^3y}{dx^3} = f'''(x),$$

and $d^3y = f'''(x) dx^3$; the fourth differential coefficient $\frac{d^4y}{dx^4} = f''''(x)$; and $d^4y = f''''(x) dx^4$; the n th differential coefficient: $\frac{d^ny}{dx^n} = f^n(x)$, and $d^ny = f^n(x) dx^n$. The Taylorian series can then, with respect to the notations here employed, be written thus:

$$f(x+h) = y + \frac{dy}{dx} \cdot \frac{h}{1} + \frac{d^2y}{dx^2} \frac{h^2}{1 \cdot 2} + \frac{d^3y}{dx^3} \frac{h^3}{1 \cdot 2 \cdot 3} + \dots$$

“For every explicitly given function of x the differential coefficient can easily be determined, and thus also the successive differential coefficients.

This act is called the differentiation of the function, and serves to find its development in a series according to increasing powers”²⁷.

If one has, then, in accordance with the foregoing (II, 1, 2, 3), the series:

$$(x+h)^n = x^n + nx^{n-1} \frac{h}{1} + n(n-1)x^{n-2} \frac{h^2}{1 \cdot 2} + \dots$$

²⁷Ramus. *Algbr.* F. S. 102.

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$$a^{x+h} = a^x a^h = a^x \left(1 + (l.a) h + (l.a)^2 \frac{h^2}{1 \cdot 2} + \dots \right)$$

$$\log(x+h) = \log x + \log \left(1 + \frac{h}{x} \right) = \log x + \log e \left(\frac{h}{x} - \frac{1}{2} \cdot \frac{h^2}{x^2} + \dots \right)$$

$$\sin(x+h) = \sin x \cos h + \cos x \sin h = \sin x + \cos x \cdot h - \dots$$

$$\cos(x+h) = \cos x \cos h - \sin x \sin h = \cos x - \sin x \cdot h - \dots$$

²⁹ one obtains, for the several series, as the first differential coefficient:

$$\begin{aligned} \frac{d.x^n}{dx} &= nx^{n-1}; & \frac{d.a^x}{dx} &= (la)a^x; & \frac{d.\log x}{dx} &= \frac{\log e}{x}; \\ \frac{d.\sin x}{dx} &= \cos x; & \frac{d.\cos x}{dx} &= -\sin x. \end{aligned}$$

But as one finds, under given presuppositions, the first, one also finds in the corresponding way the second, the third, the n th differential coefficient. Since now, for all possible functions, one can find differentials and differential coefficients of all possible orders, and since these coefficients, though different from one another, are yet, in comparison with finite magnitudes, each of them, as $\frac{0}{0}$; we have here an example which shows how the infinite, the negative, can everywhere devour and yet preserve

²⁸One has, namely:

$$\begin{aligned} f(x+h) - y &= \frac{dy}{dx} \frac{h}{1} + \frac{d^2y}{dx^2} \frac{h^2}{1 \cdot 2} \dots \\ (x+h)^n - x^n &= nx^{n-1} \frac{h}{1} + n(n-1)x^{n-2} \frac{h^2}{1 \cdot 2} \dots \\ a^{x+h} - a^x &= a^x(l.a) \frac{h}{1} + a^x(l.a)^2 \frac{h^2}{1 \cdot 2} \dots \\ \log(x+h) - \log x &= \log e \cdot \frac{1}{x} \frac{h}{1} - \log e \frac{1}{x^2} \frac{h^2}{2} \dots \end{aligned}$$

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$$\begin{aligned} \sin(x+h) - \sin x &= \cos x \frac{h}{1} - \dots \\ \cos(x+h) - \cos x &= -\sin x \frac{h}{1} + \dots \end{aligned}$$

By marking in each series that which corresponds to $\frac{dy}{dx}$, one obtains the required values.

the finite, the positive, as a moment, a co-determinant in the punctual relations, indeed in the relation of the relations.

Under these circumstances there can here no longer be talk either of a mere finite or of leap or of approach, but only of determinacy, inner infinite determinacy. How peculiar the direction of thought must become, when the finite is thus to be uninterruptedly determined by the infinite, is seen involuntarily from the principles³⁰, which first come to application in the higher mathematics. One speaks of the differential calculus as of a peculiar invention, and rightly: in the analysis of the infinite everything is new; only it must not be forgotten that the infinite has from the beginning been present in the finite, that the new principles and the new laws are consequences — be

³⁰For elucidation of the differentiation of composite functions, which as it were furnishes the intellectual types, the following is here set down. I) $day = ady$, when a is constant, for a constant magnitude naturally has no differential. II) When $y, z, u \dots$ are any functions whatever of x , $d(y + z + u \dots) = dy + dz + du \dots$ III) If one sets $y = f(x)$, $z = \varphi(x)$, $yz = F(x)$, one obtains: $F(x + h) = f(x + h) \cdot \varphi(x + h)$, and $d.yz = ydz + zdy$. IV) If one sets $y = f(x)$, $z = \varphi(x)$, $\frac{y}{z} = F(x)$, one obtains: $F(x+h) = \frac{f(x+h)}{\varphi(x+h)}$, which by series-expansion and reduction gives $F'(x) = \frac{\varphi(x) \cdot f'(x) - f(x) \cdot \varphi'(x)}{(\varphi(x))^2}$ or $d \cdot \frac{y}{z} = \frac{zdy - ydz}{z^2}$; V) If one sets $z = \varphi(x)$, $y = f(z) = F(x)$, one gets by development and reduction $F'(x) = f'(\varphi(x)) \cdot \varphi'(x)$ or: $\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx}$. VI) When $f'(x)$ can be expressed by $f(x)$, such as $f'(x) = \varphi[f(x)]$, and the inverse function of $f(x)$ is called $f_1(x)$, one has: $y = f(x)$, $x = f_1(y)$, $dy = \varphi(f(x)) dx = \varphi(y) dx$; thus $dx = df_1(y) = \frac{dy}{\varphi(y)}$ or: $d \cdot f_1(x) = \frac{dx}{\varphi(x)}$ (whose application is important when one, for example, will differentiate the arc corresponding to a trigonometric magnitude).

it noted: teleological consequences — of the old elementary presuppositions³¹.

c) The Reciprocal Relation of the Differential Coefficients.

If, with special regard to the series:

$$f(x+h) = y + \frac{dy}{dx} \frac{h}{1} + \frac{d^2y}{dx^2} \frac{h^2}{1 \cdot 2} + \frac{d^3y}{dx^3} \frac{h^3}{1 \cdot 2 \cdot 3} + \dots$$

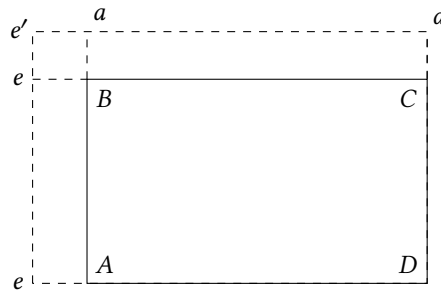
we employed the notation:

$$f(x+h) = y + \frac{0}{0} \frac{h}{1} + \frac{0^2}{0^2} \frac{h^2}{1 \cdot 2} + \frac{0^3}{0^3} \frac{h^3}{1 \cdot 2 \cdot 3} + \dots$$

then we would indeed thereby be reminded of the infinite smallness of the differentials, but would still not attain true insight into the peculiar significance of the different differential coefficients.

Let the quotient $\frac{dy}{dx} = \frac{0}{0}$ be — what, moreover, must not be taken too literally — for example $= \frac{2}{3}$; one would then be much mistaken if one concluded from this that

³¹What, for example, the formula $d.yz = ydz + zdy$ means can, with respect to the elementary presuppositions, be made intuitive:



$D = BC = ad = y$; $DC = AB = ec = z$; $Ae = ae' = dy$; $Cd = Ba = dz$. If one imagines D as the point of departure for two simultaneous movements, so that one has $DA = y = f_1(t)$, and $DC = z = f_2(t)$; then the rectangle $AD \cdot DC = yz = f_1(t) \cdot f_2(t)$. How great will the increment now be, when the movements are continued for an infinitely small time? One obtains $d(yz) = d[f_1(t) \cdot f_2(t)] = ydz + zdy = f_1(t) \cdot df_2(t) + f_2(t) \cdot df_1(t)$. The whole must then, after the differentiation has been carried out and the increment found, give $yz + d.yz$, and $d.yz = ydz + zdy$.

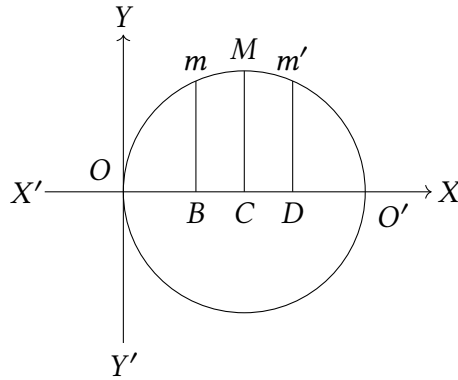
This result can be judged from the elementary standpoint. The rectangle $eDde' = (y + dy)(z + dz) = yz + ydz + zdy + dzdy$ gives, namely, 1) a product of finite with finite, yz ; 2) of finite with infinite, ydz and zdy ; 3) of infinite with infinite: $dzdy = 0 \cdot 0 = 0^2$, which drops away; in other words: $ce'ab = dzdy$ is regarded as a point, while $adaB = ydz$ and $ABBc = zdy$ are reckoned in as infinitely small rectangles.

$\frac{d^2y}{dx^2} = \frac{0^2}{0^2}$, and so must be $\left(\frac{2}{3}\right)^2$, $\frac{d^3y}{dx^3} = \left(\frac{2}{3}\right)^3$, and so forth. To be sure, $d^n y$ is in general of the same order, that is to say: comparable with dx^n , for only for special values of x does one have either $f^n(x) = 0$ or $f^n(x) = \infty$; but nevertheless the notation itself reminds one that $d^n y$ does not relate to y in the same way as dx^n relates to x .

For in $\frac{d^2y}{dx^2}$, $dx^2 = dx \cdot dx$; in $\frac{d^3y}{dx^3}$, $dx^3 = dx \cdot dx \cdot dx$, and so on; whereas, on the other hand, $d^2y = d \cdot dy$, $d^3y = d \cdot d^2y$, and so forth. With this it also agrees, as regards the matter, that $\frac{dy}{dx}$ can in certain cases be 0, while $\frac{d^2y}{dx^2}$ is a finite magnitude; which would be impossible if the differential coefficients in a given series related to one another as increasing powers of the same, proper or improper, fraction. To be sure, f''' in the series is derived from f'' in the same way as f'' is derived from f' ; but from this it by no means follows that one should have $f'' = (f')^2$, $f''' = (f')^3$.

If we call, to choose an example from rational mechanics, the velocity v , the time t , and the space traversed x ; then in a finite time $v = \frac{x}{t}$; but in the infinite time of the moment $= \frac{dx}{dt}$. If we next denote by φ the moving force, and seek the force in its measure, then for a finite time we have $\varphi = \frac{v}{t}$, but for the infinite time of the moment $\varphi = \frac{dv}{dt}$. Since now the velocity in each moment is $v = \frac{dx}{dt}$, then $dv = d \cdot \frac{dx}{dt}$; $\varphi = \frac{dv}{dt}$ then becomes $= \frac{\frac{d^2x}{dt}}{dt} = \frac{d^2x}{dt^2}$; when, then, the first differential coefficient denotes a velocity, the second differential coefficient denotes the force acting in the moment.

Another, equally comprehensible example can be drawn from maximum and minimum.



A function of x is at its maximum when $f(x + h)$ and $f(x - h)$ are both less than

$f(x)$, or, what is the same, when $f(x+h) - f(x)$ and $f(x-h) - f(x)$ are both negative; if they are both positive, the function is on the contrary at its minimum. If $OC = x$, $BC = CD = h$, then $Dm' = f(x+h)$, $MC = f(x)$, and $Bm = f(x-h)$.

$$\begin{aligned} f(x+h) &= f(x) + f'(x) \frac{h}{1} + f''(x) \frac{h^2}{1 \cdot 2} + \dots \\ f(x-h) &= f(x) - f'(x) \frac{h}{1} + f''(x) \frac{h^2}{1 \cdot 2} - \dots \end{aligned}$$

Thus:

$$\begin{aligned} f(x+h) - f(x) &= +f'(x) \frac{h}{1} + f''(x) \frac{h^2}{1 \cdot 2} + \dots \\ f(x-h) - f(x) &= -f'(x) \frac{h}{1} + f''(x) \frac{h^2}{1 \cdot 2} - \dots \end{aligned}$$

In order that both series may have the same sign, the first term, which has opposite signs, and which, when h is very small, determines the sign of the whole series, must drop away, i.e. $f'(x) = \frac{dy}{dx} = 0$, and $f''(x) = \frac{d^2y}{dx^2}$ be negative. For a minimum $\frac{dy}{dx} = 0$ must likewise drop away, but $\frac{d^2y}{dx^2}$ on the contrary be positive.

We can make this relation intuitive. According to the elementary presentation, one has, for the circle, when in the above figure one sets $OB = x$, $Bm = y$, $OO' = 2r$, $x : y = y : 2r - x$; $y^2 = 2rx - x^2$; $y = \pm \sqrt{2rx - x^2}$. If $x = r$, then $y = CM$, and at its maximum. According to the higher analysis one has, since $y = f(x) = \pm \sqrt{2rx - x^2}$,³²

$$\frac{dy}{dx} = \pm \frac{r-x}{\sqrt{2rx-x^2}}, \quad \text{and} \quad \frac{d^2y}{dx^2} = \pm \frac{-\sqrt{2rx-x^2} - (r-x) \frac{r-x}{\sqrt{2rx-x^2}}}{2rx-x^2}.$$

For every infinitely small increment to x , y receives its likewise infinitely small increment: the values cited express the relation between the infinitely small increments.

³² $df(x) = d(\pm \sqrt{2rx - x^2})$ then becomes $= d(2rx - x^2)^{\frac{1}{2}}$. In order to carry out this differentiation, the formula $d(x^n) = nx^{n-1}dx$ is presupposed, according to which $d\sqrt{x} = d(x^{\frac{1}{2}}) = \frac{1}{2}x^{\frac{1}{2}-1}dx = \frac{1}{2}x^{-\frac{1}{2}}dx$; but $x^{-\frac{1}{2}} = \frac{1}{\sqrt{x}}$; thus $\frac{1}{2}x^{-\frac{1}{2}}dx = \frac{\frac{1}{2}dx}{\sqrt{x}}$. — According to this formula, $d(2rx - x^2)^{\frac{1}{2}} = \frac{\frac{1}{2}d(2rx - x^2)}{\sqrt{2rx - x^2}}$. — To differentiate the numerator, the formula $d(y \pm z) = dy \pm dz$ is applied; $\frac{1}{2}d(2rx - x^2)$ thus becomes $\frac{1}{2}(d2rx - dx^2)$. But $d2rx = 2r dx$, and $\frac{2r dx}{dx} = 2r$; $\frac{d(x^2)}{dx} = 2x^{2-1} = 2x$. Thus:

From the figure it is seen, what indeed also lies in the nature of the matter, that every increment for y must be positive, until y reaches its maximum, thereafter negative, and so forth.

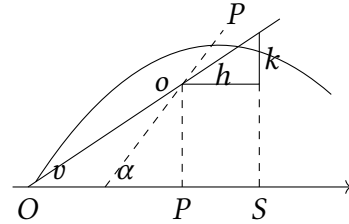
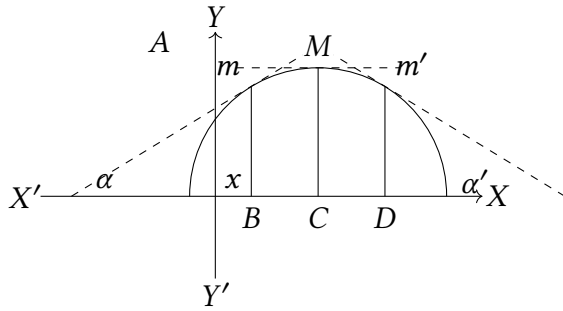
The condition for y to be at its maximum is, in the present case: $x = r$. The moment this condition is fulfilled, $\frac{dy}{dx} = \pm \frac{r-x}{\sqrt{2rx-x^2}} = 0$, since one gets $r-x=0$; on the other hand,

$$\frac{d^2y}{dx^2} = \pm \frac{-\sqrt{2rx-x^2} - (r-x) \frac{r-x}{\sqrt{2rx-x^2}}}{2rx-x^2}$$

will not become 0, but by reduction $\mp \frac{1}{r}$.

To make clear to oneself the inner connection of the matter, let one look at the tangent.

The tangent to a maximum point must be parallel with the axis of abscissas; for every other point the tangent will form an angle with the positive direction of the axis of the x 's.



$OP = x$; $oP = y$; The point $(x, y) = o$.

$$\frac{d(\pm \sqrt{2rx-x^2})}{dx} = \pm \frac{\frac{1}{2}(2r-2x)}{\sqrt{2rx-x^2}} = \pm \frac{r-x}{\sqrt{2rx-x^2}} = \frac{dy}{dx}. \text{ To find the second differential coefficient}$$

$$\frac{d\left(\frac{r-x}{\sqrt{2rx-x^2}}\right)}{dx}, \text{ the formula } d\left(\frac{y}{z}\right) = \frac{z dy - y dz}{z^2} \text{ is presupposed; one then obtains } (y = r-x;$$

$$z = \sqrt{2rx-x^2}) \frac{\sqrt{2rx-x^2} d(r-x) - (r-x) d\sqrt{2rx-x^2}}{(\sqrt{2rx-x^2})^2}. \text{ But } \frac{d(r-x)}{dx} = \frac{dr}{dx} - \frac{dx}{dx} = 0 - 1,$$

$$\frac{d\sqrt{2rx-x^2}}{dx} = \frac{r-x}{\sqrt{2rx-x^2}}, \text{ and the whole result: } \pm \frac{-\sqrt{2rx-x^2} - (r-x) \frac{r-x}{\sqrt{2rx-x^2}}}{2rx-x^2} = \frac{d^2y}{dx^2}.$$

But in order to determine the tangent, one must know its relation to the secant. According to figure B, $\frac{k}{h} = \operatorname{tg} v$, and v the angle which the secant forms with the positive direction of the axis of the x 's. As the points of intersection, p and o , move nearer to one another, the secant approaches becoming "touching", and, the moment the points coincide, will be a tangent. Since now $\frac{k}{h} = \operatorname{tg} v$, then $\lim \frac{k}{h} = \frac{dy}{dx}$ must be $\operatorname{tg} \alpha$, α being the angle which the tangent to the point (x, y) forms with the axis of the x 's.

From figure A one sees that $\operatorname{tg} \alpha$ passes, at the maximum point, over from being positive to being negative = $\operatorname{tg} \alpha'$; $\frac{dy}{dx}$ thus changes sign at M , and must then at this point be 0.

Since, further,

$$\frac{d^2y}{dx^2} = \frac{d \cdot \frac{dy}{dx}}{dx} = \frac{d \operatorname{tg} \alpha}{dx},$$

$\frac{d^2y}{dx^2}$ denotes the increment by which $\operatorname{tg} \alpha$ varies when x grows. But $\operatorname{tg} \alpha$ (to the point m) is evidently, when x varies from B to D , in constant decrease, so that the increment is negative; $\frac{d^2y}{dx^2}$ must, then, with respect to the maximum, be negative, and the formula $\mp \frac{1}{r}$ give $-\frac{1}{r}$.

In this way, then, each differential coefficient can, in accordance with the systematic connection of the derived functions, come to play a peculiar role.

3) The Integral.

An analysis of the infinite is necessarily at the same time an analysis of the finite, just as also every analysis of the negative is at the same time an analysis of the positive, every analysis of concept likewise an analysis of representation.

In relation to the line as the positive, the point, the limit of the line, is the negative; when the line is a representation, the point is a thought. In relation to the surface as the positive, the line, the limit of the surface, is the negative: the surface is represented when the line is thought. In relation to the body, the positive, the surface, the limit of the body, is the negative.

In relation to the still undetermined place, the negative, the determinate point is the positive; indeed, in relation to the mass, the material object, the mathematical

body itself is a negation, an object of thought, a concept.

The infinite and yet withal finite reciprocal determination of negative and positive, of point and line, of line and surface, of surface and body, which occupies mathematics, becomes then also recognizable by the categories: discreteness and continuity.

A line, this given continuum, can be divided, and each part is again a line. Since now each part becomes the smaller, the greater the number of parts becomes, one concludes that every line must be able to be divided, or rather: “be thought as divided into infinitely many infinitely small parts”.

The inference leads, however, at once into several difficulties. If each smallest part of the divided line is still a line, a little continuum, then this continuum must again be able to be divided into infinitely many infinitely small parts.

The smallest part is then so far from being the smallest possible, that on the contrary it even seems to become equal in size to the whole line, since the part and the whole both admit of being divided into “infinitely many infinitely small parts”. If, on the other hand, each smallest part of the divided line can no longer be divided, then this smallest part is no more a line but a point.

The divided line then comes to consist in an infinite discreteness of infinitely many points; but the point is precisely the negation of the line, is, punctually understood, without any extension, a Not-Line. According to the cited inference the line came then to consist, not of countless small lines, but of infinitely many Not-Lines. But to imagine a real, positive line composed of infinitely many Not-Lines is about as difficult as to imagine a real, positive Thanks composed of infinitely many Un-thanks.

What mathematics really puts into the expression “infinitely many infinitely small parts” cannot, however, be known from turns of phrase; it must show itself in the deed. If we seek the infinite smallness of the parts in the differential, we must seek their so-called infinite sum in the integral; from summation and integration we shall then strive to find out the determination of the integral.

a) Summation.

If a line x is assumed to be divided into infinitely many infinitely small parts, and each of these parts — without its yet being decided whether they are points or lines —

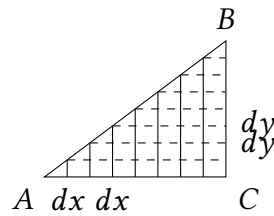
is denoted by dx , then mathematics is incontestably right when it denotes the whole line by the infinite sum of the parts $\int dx = x$.

But what now is dx , point or line? Should dialectics explain itself on this, it would say: it is not a point, for it is really a line; it is not a line, for it is essentially a point. It is a line which has just shrunk to a point; a point which is just in the process of being expanded into a line.

What dialectics thus says in its language is precisely what mathematics presupposes in its practice. dx is, as already shown, just as little 0 as a finite magnitude. If dx were 0, then $\int dx$ would simply be $\infty \cdot 0$, but this is by no means the case. If one sets $\int dx = \infty \cdot 0 = \frac{1}{0} \cdot 0 = \frac{0}{0}$, then there is here no difference between the integral and a still undetermined differential coefficient, whose value can thus be both finite and infinite.

Mathematics, then, does not engage in determining dx immediately, be it as point or as line; dx is determined only in a relation, as, for example, by dy ; so that the law of change can show itself in $f(x) = y$.

In $f(x) = y$ the increment for x , the independent, must naturally be constant; for if dx varied, the changes would have to be subject to a certain law, and x could then not be independent. As regards the increment for y , the dependent, the differential coefficient $\frac{dy}{dx}$ expresses a definite relation, but the individual dy can, on the other hand, be among themselves either equal or unequal.



$$AC = x; \quad BC = y.$$

In the triangle ABC one has:

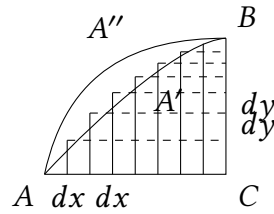
$$BC = y = dy + dy + dy + \dots$$

$$AC = x = dx + dx + dx + \dots$$

In each of these infinite sums the terms are among themselves equal, namely $dy = dy$, just as $dx = dx$; but only in a single case, namely when $\angle BAC = 45^\circ$, is $dy = dx$, and the differential quotient $\frac{dy}{dx} = \frac{0}{0} = 1$.

In the figures $AA'BC$ and $AA''BC$ one still has, as before, $dx = dx$, but on the other hand not $dy = dy^1$. If the dependence in which y stands to x is viewed, for one case by the curve $AA'B$, for another by the curve $AA''B$; then it is evident from this that the individual dy in the infinite sum:

$$y = dy + dy' + dy'' + dy''' + \dots$$



$$AC = x; \quad BC = y.$$

will not only be, in both cases, among themselves unequal, but the inequality withal also different for every one of the different cases.

If one now holds fast the presupposition that every mathematical point must, as negation of all magnitude, be 0, so that $0 = 0$ in this case is as Nothing = Nothing; then it is clear that dx in the line x can just as little be a mathematical point as dy in the line y , for from Nothing comes Nothing, and between Nothing and Nothing there is no relation. But that dx and dy are nevertheless not restricted to being lines of finite extension is seen also from this, that the law of dependence for y with respect to x is each time so determined that the relation $\frac{dy}{dx}$ is preserved even in $\frac{0}{0}$.

If the differential denotes a moment in the infinite discreteness of the line, then the integral denotes the merging of the discrete moments into the finite continuum; if the differential denotes the infinitely small term, then the integral denotes the infinitely great sum. If the infinitely small term, be it dx or dy , cannot be directly stated, then the infinite summation, the summation of the countless terms, cannot be directly carried out either.

If one presupposes $f(a + dx) - f(a) = f'(a) dx$, one obtains, by constantly adding dx ,

$$\begin{aligned} f(a + 2dx) - f(a + dx) &= f'(a + dx) dx, \\ f(a + 3dx) - f(a + 2dx) &= f'(a + 2dx) dx, \\ &\dots\dots\dots \\ f(a + ndx) - f(a + (n-1)dx) &= f'(a + (n-1)dx) dx \end{aligned}$$

and by addition:

$$f(a + ndx) - f(a) = f'(a) dx + \dots f'(a + (n-1)dx) dx.$$

If dx is infinitely small, $f'(a) dx$, $f'(a + dx) dx \dots$ likewise, one must set $n = \infty$, in order that ndx may become a finite magnitude, and $a + ndx$ in $f(a + ndx)$ is set $= b$.

But when the matter is thus conceived, the old difficulties return; it is only by a leap that one can here reach from n to ∞ , and when one has then — naturally by leaping over infinitely many terms — made this leap, the sum, ∞dx , is still so indeterminate a magnitude that it can be set $= x$, $= \frac{1}{2}x = 1000x$, $= \frac{1}{1000}x$, and so on.

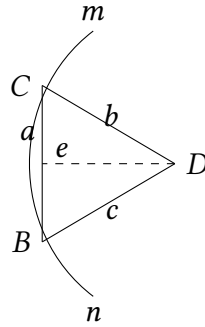
A sum is, according to its concept, a number, a finite number. An infinite number, an infinite sum, a summation of an infinite number are at bottom nothing but contradictions. If an infinite summation is to be carried out, the task must be solved either by approach, thus in approximateness, or by a transformation.

b) Integration.

In the series $1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \dots$ the number of terms is infinite. A summation of these terms can now, in the first place, take place by approach, in that, with regard to the convergence of the series, one takes as many terms as the accuracy of the calculation requires; but without losing oneself in a counting-together of countless terms, one could also by another way attain a result, and even an exact result, namely by finding the quotient from which the infinite series springs. Instead of an infinite summation, a law-governed reduction is then employed: the task is solved by being transformed.

A telling example of the transformation of tasks we have in the treatment of the relation of broken lines to curves.

At the transition from finite to infinite, mathematics begins with the assumption that the circle is a polygon of an infinite number of infinitely small sides. How this proposition is applied in elementary geometry is well known; in the higher analysis the matter can be presented thus:



$a = b \cos C + c \cos B$, according to the figure; but

$$\cos C = 1 - \frac{C^2}{1 \cdot 2} + \frac{C^4}{1 \cdot 2 \cdot 3 \cdot 4} \dots,$$

and

$$\cos B = 1 - \frac{B^2}{1 \cdot 2} + \frac{B^4}{1 \cdot 2 \cdot 3 \cdot 4} \dots$$

Thus

$$\begin{aligned} a &= b \left(1 - \frac{C^2}{1 \cdot 2} \dots \right) + c \left(1 - \frac{B^2}{1 \cdot 2} \dots \right) = b - \frac{bC^2}{1 \cdot 2} \dots \\ &\quad + c - \frac{cB^2}{1 \cdot 2} \dots = b + c - \frac{bC^2}{1 \cdot 2} \dots - \frac{cB^2}{1 \cdot 2} \dots \end{aligned}$$

The fundamental question is what, according to the stated law of change, will happen when the terms become infinitely small; and the answer is that $\frac{bC^2}{1 \cdot 2}$ and $\frac{cB^2}{1 \cdot 2}$, the terms of the second order, will vanish, and leave $a = b + c$.

In the finite sense the arc CB corresponding to a as chord is $< b + c$, but $> a$; but, since $a = b + c$, the whole BCD has shrunk to a point, which is nevertheless not a point, but an element in a line.

The line in which BCD is an element is not a polygon-side, but a circle-periphery.

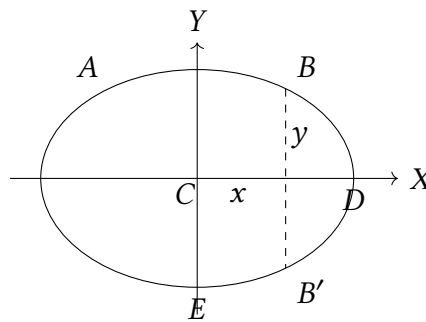
In so far as there is here in *a* still the least remnant of a polygon-side, *BCD* is not an element in a line, but a real triangle; the triangle *BCD* can vanish, but only in the curved line.

Instead of proving either that the circle is a polygon or the polygon a circle, mathematics has on the contrary transformed the task, and proved that a circle is essentially different from a polygon, and a polygon from a circle.

The view that a curved line is at bottom a broken line is intimately connected with the conception that integration is a summation. Instead of conceiving integration as an infinite summation, we must rather regard it as a transformation of the task of summation, more closely determined by the express opposition to differentiation³³. Leibnitz, who discovered differentiation, as the direct tangent method, found in integration the inverse tangent method.

With respect to the circle, $y = f(x) = \pm \sqrt{2rx - x^2}$ was in the foregoing given as the original function; the task was then to find the derived functions, namely $\frac{dy}{dx} = \frac{r - x}{\sqrt{2rx - x^2}}$, and thereby to determine the angle which the tangent to a point in the circle will form with the positive direction of the axis of abscissas. By inserting different values for *x* one determines the tangent for the different points in the curve.

If we choose the ellipse, and place the origin of the coordinates at the centre, then we again have $y = f(x)$, but this time determined as $\frac{b}{a} \sqrt{a^2 - x^2}$.



^{33a}Integration is the inverse calculation of differentiation. It consists, when a differential is given, in determining the function of which it is the differential, and in general, when a differential equation is given, in determining its corresponding primitive equation". Ramus. Differ. Integr. S. 34. That the higher theories of the summation of infinite series rest on reductions and transformations is evident. A series such as the Eulerian: $x^{-a_m-1} \int_0^x s_{n,m} x^{a_m} dx = s_{n,m-1}$. (Ramus. Diff. Integr. S. 308) admits, naturally, still less of being treated by direct summation than, for example, the series: $1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \dots$

$$CD = a; \quad CE = b.$$

The first differential coefficient corresponding to this is then found to be $-\frac{b}{a} \frac{x}{\sqrt{a^2 - x^2}}$, whereby likewise the direction of the tangent for every point in the ellipse can be determined. If, for example, $x = 0$, one gets the direction of the tangent at the end-point of the minor axis; for then

$$\frac{dy}{dx} = -\frac{b}{a} \frac{0}{\sqrt{a^2 - 0}} = 0.$$

If $x = a$, one gets the tangent to the end-point of the major axis, namely

$$\frac{dy}{dx} = -\frac{b}{a} \frac{a}{\sqrt{a^2 - a^2}} = -\frac{ba}{\sqrt{0}} = -\frac{ba}{0} = \infty.^{34}$$

In both these cases, then, the original function $f(x)$ was given, and the task was to find the derived one,

$$\frac{dy}{dx} = \frac{d f(x)}{dx}.$$

These, then, were examples of the direct tangent method.

But now, conversely, a derived function $f'(x) = \frac{dy}{dx}$ could also be given, and the task become to find the corresponding original one, $f(x) = y$.

1) Example. There is given an equation $\frac{dy}{dx} = \frac{r-x}{\sqrt{2rx-x^2}} = f'(x)$, whereby a curve's tangent is determined; the task

$$\text{A series such as } \int_0^x 2x \, dx = \text{Lim } 2 \frac{x^2}{n^2} (1 + 2 + 3 + 4 + \dots (n-1))$$

is not summed, but reduced:

$$= \text{Lim } 2 \frac{x^2}{n^2} \left(\frac{(n-1)n}{2} \right) = \text{Lim } x^2 \frac{n-1}{n} = \text{Lim } x^2 \left(1 - \frac{1}{n} \right).$$

From the result of the reductions it emerges that the sum of the series, when one sets $n = \infty$, will give x^2 , and that one thus obtains $\int_0^x 2x \, dx = x^2$; but such a summation

³⁴To AB, the tangent, which forms an angle = 0 with the axis of the x 's, there corresponds a tangent (in numerical value) = 0; to $A'B'$, a tangent which forms an angle with the axis of the x 's = 90° , there corresponds a tangent = ∞ .

is no longer summation: the limit is transgressed, the task transformed. is, to find the curve's original equation: $f(x) = y$. By integration, the inverse tangent method, one then finds

$$\int f'(x) dx = f(x) = \pm \sqrt{2rx - x^2};$$

i.e. one finds that the curve in question is a circle.

2) Example. There is given an equation $\frac{dy}{dx} = -\frac{b}{a} \frac{x}{\sqrt{a^2 - x^2}}$, whereby the tangent for a curve is determined; one is now, by integration, the inverse tangent method, to find what kind of curve it is, and then finds $\int f'(x) dx = \frac{b}{a} \sqrt{a^2 - x^2} = f(x)$, which shows that the curve is an ellipse.

The opposition between integration and differentiation can, with respect to powers, be made intuitive thus: by differentiation one passes from $f(x) = x^n$ to $f'(x) = nx^{n-1}$; by integration one passes conversely from $f'(x) dx = nx^{n-1} dx$ to $f(x) = x^n$: by differentiation one has $d(x^n) = nx^{n-1} dx$, by integration one obtains $\int nx^{n-1} dx = \int d(x^n) = x^n$; one differentiates a power by multiplying by the exponent and subtracting 1 from the exponent; one integrates a power by adding 1 to the exponent and dividing by the increased exponent.

$$x^n = \int nx^{n-1} dx = n \int x^{n-1} dx, \quad \text{thus:} \quad \frac{x^n}{n} = \int x^{n-1}$$

dx , if one now sets $n - 1 = m$, $n = m + 1$, then one has:

$$\int x^m dx = \frac{x^{m+1}}{m+1} + C.$$

But what now does C , the arbitrary constant, mean?

c) The Determination of the Integral.³⁵

³⁵There is no general method for finding $\int X dx$ expressed under finite form, when X is an explicit function of x under finite form, whereas $\frac{dX}{dx}$ can always be expressed under finite form. For integration is an inverse calculation, and therefore generally leads to new functions"... "Just as subtraction, division, root-extraction, which are inverse calculations of addition, multiplication, raising to a power, have led to new magnitudes: negative, fractional, rational, and imaginary." Ramus. Differential- og Integral-Rgn. S. 36.

In opposition to the differential as the negative, the integral is the positive. If the differential is a point, then the integral is a line; if the differential is a line, then the integral is a surface; if the differential is a surface, then the integral is a body.

The negative cannot, however, of itself produce the positive, just as little as the immediately positive can produce the negative: the transition from negative to positive, and conversely, presupposes a positing and negating activity, a principle.

In the negating activity the positive is presupposed; it is the positive that in the negation is, so to speak, cut through, cut across, or cut off by the negative: the positive is divided. In the positing activity, conversely, the negative is presupposed. It is not the being, the positive, that is posited, for the being already is, the positive is already posited; no, it is precisely the Not-Being, that which is not yet, the negative, that is

That integration, moreover, is on the whole determined by its principled opposition to differentiation lies in the nature of the matter, and is seen from manifold forms. Since a constant can have no differential, it can have no integral either. When a is constant, $\int a X dx = a \int X dx$. If $d(\sin x) = \cos x dx$, then $\int \cos x dx = \int d(\sin x) = \sin x$; if $d(\cos x) = -\sin x dx$, then $\int -\sin x dx = \int d(\cos x) = \cos x$, and $-\cos x = \int \sin x dx$; if $d(f(x) \pm \Phi(x)) = f'(x) dx \pm \Phi'(x) dx$, then $\int (f'(x) dx \pm \Phi'(x) dx) = \int d(f(x) \pm \Phi(x)) = f(x) \pm \Phi(x)$; if $d(f(x) \cdot \Phi(x)) = f(x) d\Phi(x) + \Phi(x) df(x)$, and $f(x) d\Phi(x) = d(f(x)\Phi(x)) - \Phi(x) df(x)$, then $\int d(f(x) \cdot \Phi(x)) = f(x)\Phi(x) = \int f(x) d\Phi(x) + \int \Phi(x) df(x)$, and $\int f(x) d\Phi(x) = f(x)\Phi(x) - \int \Phi(x) df(x)$. But in the further development the matter becomes complicated; already $\int \frac{f(x)}{\Phi(x)}$ requires more elaborate operations:

$$\frac{f(x)}{\Phi(x)} = F(x) + \frac{k_1}{x - \alpha_1} + \frac{k_2}{x - \alpha_2}.$$

Set

$$\frac{1}{x^2 - 7x + 12} = \frac{1}{(x - 4)(x - 3)} = \frac{k_1}{x - 4} + \frac{k_2}{x - 3}; = \frac{k_1(x - 3) + k_2(x - 4)}{(x - 4)(x - 3)};$$

thus $1 = (k_1 + k_2)x - (3k_1 + 4k_2)$. By comparing the powers of x on both sides it is seen that $k_1 + k_2 = 0$, and $-3k_1 - 4k_2 = 1$; $4k_1 + 4k_2 = 0$; $k_1 = 1$; $k_2 = -1$. Consequently:

$$\frac{1}{x^2 - 7x + 12} = \frac{1}{x - 4} - \frac{1}{x - 3};$$

that is:

$$\int \frac{dx}{x^2 - 7x + 12} = \int \frac{dx}{x - 4} - \int \frac{dx}{x - 3}.$$

For $\Phi(x) = (x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_p)^n (x - \alpha)^m \dots$ one has

$$\int \frac{1 \cdot dx}{x - 4}, \quad \int \frac{3 dx}{(x - 4)^2}, \quad \int \frac{A dx}{(x - \alpha)^m}.$$

Ifr. Ramus. Diff. Integr. S. 37.

posited, and transposed. That the point, the negative, is posited positively means, in other words, that the beginning of a line is posited; as the posited beginning is continued, the line comes to be: the continuation is a movement.

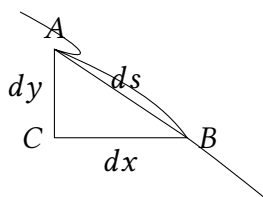
That in this connection the categories division and composition do not suffice, but that division and movement ought to be opposed to one another, can be seen from a comparison between the broken and the curved line.

For the broken line a composition holds, for its parts are finite; it is a piecework which in each and all belongs to finitude. For the curved line no finite composition whatever holds, not even the very finest piecing-together; it owes its coming-into-being to the movement alone, for it belongs to infinity.

The so-called composite movement is not a piecing-together of two movements, but a movement mediated through two moments. In so far as there is here the least trace of real piecing-together, the moved body will change its direction only at finite intervals: there will arise a broken, not a curved line.

The principle of movement which produces the curved line is intimately akin to the nature of the function; here is continuity in infinite discreteness, an invariable in infinite change.

But on the function both the differential and the integral depend.



$A = ds$, the infinitely small sector.

For any curve whatever one has

$$ds^2 = dx^2 + dy^2; \quad ds^2 = dx^2 \left(1 + \frac{dy^2}{dx^2} \right); \quad \frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx} \right)^2}.$$

This relation is each time peculiarly determined by the curve's equation; $f(x) = y = \sqrt{a^2 - x^2}$ for the circle; $\frac{b}{a} \sqrt{a^2 - x^2}$ for the ellipse; $\frac{b}{a} \sqrt{x^2 - a^2}$ for the hyperbola;

$\sqrt{x} \sqrt{p}$ for the parabola³⁶, o. s. v.

What is here indicated by the differential coefficient is, then, not an already existing fragment, but a moment which has significance only in relation to the definite law of movement, and the movement therewith either already entered upon or at least entering.

If the expression $-\frac{b}{a} \frac{x}{\sqrt{a^2 - x^2}} = \frac{dx}{dy}$ for the ellipse is inserted, one obtains by reduction

$$\frac{ds}{dx} = \sqrt{\frac{a^4 + (b^2 - a^2)x^2}{a^2(a^2 - x^2)}},$$

thus

$$ds = dx \sqrt{\frac{a^4 + (b^2 - a^2)x^2}{a^4 - a^2x^2}},$$

a formula which is especially used to rectify the ellipse, or to present it by quadrature³⁷, in that by integrating one obtains:

$$s = \int_0^{x=a} dx \sqrt{\frac{a^4 + (b^2 - a^2)x^2}{a^4 - a^2x^2}},$$

a transformation which cannot possibly be brought about by piecing-together, but is due solely to the movement and the law holding for the movement.

Even as regards the incommensurable, there can here be no talk of inexactness, but only of an inner endowment for infinite advance, infinite development, infinite movement.

From division and movement, then, the opposition between the differential and the integral in calculations of areas and volumes is thus understood.

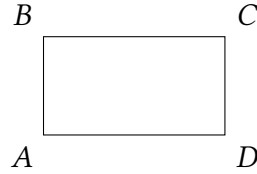
That a surface can be divided into infinitely many infinitely small surfaces, is incontestable; but that a finite surface cannot be pieced together from infinitely many infinitely small surfaces, is also incontestable. Let $\int y dx = A$ be the typical formula for any area whatever, $y = f'(x)$.

³⁶Ifr. Ramus. Analytisk Geometrie. S. 13–23.

³⁷Every differential equation which has been solved in such a way that the one variable has been expressed by an integral which contains the other alone is said to be presented by quadrature. On quadrature and rectification cf. Ramus. Analytisk Geometrie. S. 112–125.

1) Example. In the rectangle yx , $y = a$, $ydx = adx$. According to the formula:

$$\int y dx = \int a dx = adx + adx + adx + \dots$$



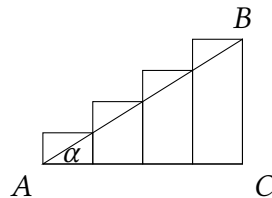
$$AD = x; \quad DC = y.$$

$$AB = a.$$

one might almost suppose that mathematics really intended to count together the countless adx , and to piece the finite rectangle together of infinitely many infinitely small rectangles. This is, however, a misunderstanding. The task is transformed by being solved. In $\int y dx = \int a dx$, $a = y$ is constant; thus $\int a dx = a \int dx = ax$: the rectangle = base $\times$ height.

The solution of the task here by no means aims at showing how the finite is pieced together from the infinite, but at finding the finite corresponding to the infinite, namely the yx which admits of being divided into the countless ydx . Should the presentation show how yx arises from ydx , then it would have to demonstrate the movement expressly.

2) Example. In the triangle ABC one has $\int y dx = ydx + y'dx + y''dx + \dots$; but instead of piecing the triangle together of infinitely many infinitely small rectangles, mathematics conceives y as a function of x ; $y = \operatorname{tg} \alpha x$, thus $ydx = \operatorname{tg} \alpha x dx$, and this conception corresponds precisely to the movement.



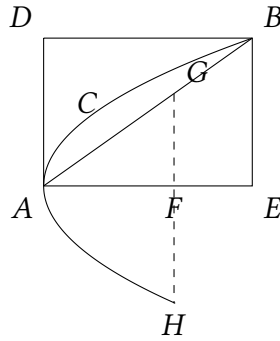
$$AC = x; \quad CB = y.$$

One thus obtains $\int y dx = \int \operatorname{tg} \alpha x dx = \operatorname{tg} \alpha \int x dx$; but according to the law

stated for the integration of a power, $\int x dx = \frac{x^2}{2}$, and thus

$$\operatorname{tg} \alpha \int x dx = \operatorname{tg} \alpha \frac{x^2}{2} = \operatorname{tg} \alpha x \frac{x}{2} = y \frac{x}{2} :$$

the area of the triangle = base $\times \frac{1}{2}$ height.



$$AE = x; \quad EB = y.$$

$$GFH = p.$$

For a parabola, whose equation, $y^2 = px$, shows $y = f'(x) = \sqrt{p} \sqrt{x}$,

$$\int y dx = \sqrt{p} \int \sqrt{x} \cdot dx = \sqrt{p} \int x^{\frac{1}{2}} dx = \sqrt{p} \frac{x^{\frac{3}{2}}}{\frac{3}{2}} = \frac{2}{3} \sqrt{p} x^{\frac{1}{2}} \cdot x = \frac{2yx}{3}.$$

$$\text{The segment } ACB = \frac{2yx}{3} - \frac{yx}{2} = \left(\frac{4}{6} - \frac{3}{6} \right) xy = \frac{1}{6} xy.$$

$$\text{The area } ADBCA = xy - \frac{2}{3} xy = \frac{1}{3} xy = 2 \times \text{the segment } ACB.^{38}$$

In all such cases that which has come to be is treated not as an immediate being, but according to its coming-into-being; the function expresses the law of the coming-into-being.

The coming-into-being here too presupposes something already come-to-be. In that integration, on the presupposition of the movement expressed in the changes, leads the infinite back to finite determinacy, the definite integral must be stated by

³⁸We have found the area $ABCE = \frac{2}{3} xy = \frac{2}{3}$ of a rectangle, which can be reduced to a square; the parabola can therefore, what the other conic sections cannot, be squared under finite form.

an added constant. The constant is precisely the limit in which the finite touches the infinite in the movement³⁹.

Example. From rational mechanics the relation between the velocity $v = \frac{dx}{dt}$, and the moving force $\varphi = \frac{d^2x}{dt^2}$, has already been brought out.

If now 1) the velocity is constant, then one sets $\frac{dx}{dt} = a$, where a is the constant. From $dx = a dt$ one then gets by integration $\int dx = \int a dt = a \int dt$, or: $x = at + b$; b is an added arbitrary constant, for if one differentiates $x = at + b$, one obtains again $dx = d \cdot at + db = a dt + 0$. The meaning of the added constant is then elucidated thus: for $t = 0$, $x = b$, thus b is the distance of the mass from the point of departure at the beginning of the movement.

If 2) the force φ is constant, $= k$, one has $\frac{d^2x}{dt^2} = k$, and $\frac{d^2x}{dt^2} = k dt$, which, integrated, gives $\frac{dx}{dt} = \int k dt$; thus $\frac{dx}{dt} = kt + c$, and c , the constant, an initial velocity, for, when $t = 0$, $\frac{dx}{dt} = c$. Integrating again, one obtains

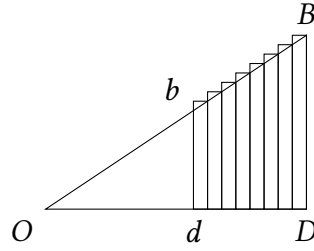
$$\int dx = \int (kt + c) dt = \int (ktdt + cdt) = k \int t dt + c \int dt,$$

thus: $x = \frac{1}{2}kt^2 + ct + d$, where d is a new constant, and so forth.

The meaning of the constant shines forth likewise from the calculation of area.

If one determines the area of the triangle by the formula $y \frac{x}{2}$ from the standpoint of finitude, then from the standpoint of infinity one must apply the formula $\int y dx = \frac{xy}{2} + C$. Here it is presupposed that x , the independent, and y , the dependent, vary to infinity: the added constant points to the movement.

³⁹The doctrine of the constant must not be misunderstood, as if mathematics, in order to indicate definite relations and express the definite laws, necessarily had to have individual invariable terms; on the contrary, the deeper one penetrates into the higher analysis — into differentials with several independents (Ramus. Diff. Integr. S. 149 and following), into the generation of curves and surfaces (Ramus. Analytisk Geometrie. S. 135 and following), into the contact, osculation, evolution of lines, and so forth — the more evident it becomes that every term of relation is infinitely variable, and only the system of laws the essential, the constant, the eternal. In so far as materialism, with respect to the relations exact in mechanical and chemical attractions, would infer from the constancy of the relations to the invariability of the fundamental parts by appealing to the mathematical, the inference is false.



$$OD = x, BD = y, Od = x' = bd = y'.$$

$\int y dx = \int f'(x) dx = f(x) + C$. If $x = 0$, then $f(0) = 0$; $\int y dx = 0 + C$, and $C = 0$; if $x = OD$, then

$$\int_0^x y dx = \int_0^x f'(x) dx = f(x);$$

if $x = Od = x'$, then

$$\int_0^{x'} y dx = \int_0^{x'} f'(x) dx = f(x').$$

In so far as the varying x and x' are in a certain relation to one another, the varying areas OBD and Obd are in a corresponding relation. This relation is determined by the area-difference, and by this in turn the constant.

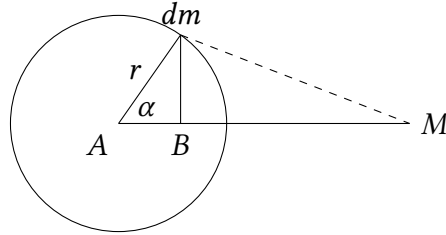
$$OBD - Obd = bdBD = \int_0^x y dx - \int_0^{x'} y dx = \int_{x'}^x y dx = f(x) - f(x').$$

In order now to find out whether the constant is positive or negative, one must begin by letting the area-difference shrink to 0. This happens in the present case by setting $x = x'$, whereby one obtains a definite limit, either bd or BD .

The area $bdBD$, which before was $= f(x) - f(x')$, now becomes $= f(x') - f(x') = f(x') + C = 0$, and consequently $C = -f(x')$. When now x , by the movement of the end-point, i.e. by continuous increment dx , is extended beyond x' , the area-difference will become $= f(x) + C$, since $C = -f(x')$. One will thus, however much x grows, always be able to set

$$bdBD = \int_{x'}^x y dx = f(x) - f(x') = f(x) + C.$$

How the transition from point to line, from line to surface, from surface to body relates to the integral is seen from the manner in which one integrates with respect to mass.



$$AM = a; \quad AB = r \cos \alpha.$$

If one assumes a corporeal point, a material element dm , which attracts M , and this dm determined by the radius r , by the angle α , and by an angle Θ which a fixed plane forms, for example, with the plane of the paper; if one further assumes $dm = p dw$, p being the density and w the volume, so that dw is proved to be $= r^2 \sin \Theta dr d\Theta d\alpha$, and the whole attraction of the point, resolved along the direction toward the centre,

$$= \frac{kdm(a - r \cos \alpha)}{(r^2 + a^2 - 2ra \cos \alpha)^{\frac{3}{2}}}$$

then one can by integration determine the attraction which the whole mass of the sphere exerts upon M ,

$$= k \int \frac{dm(a - r \cos \alpha)}{(r^2 + a^2 - 2ra \cos \alpha)^{\frac{3}{2}}};$$

but $dm = p dw$, thus one has

$$kp \int \frac{dw(a - r \cos \alpha)}{(r^2 + a^2 - 2ra \cos \alpha)^{\frac{3}{2}}} = kp \int \frac{r^2 \sin \Theta dr d\Theta d\alpha (a - r \cos \alpha)}{(r^2 + a^2 - 2ra \cos \alpha)^{\frac{3}{2}}}.$$

But to integrate once with respect to mass is the same as to integrate 3 times with respect to the variables by which the volume is expressed. Thus one integrates first with respect to Θ , the angle which arises as dm is turned in a circle about AM (the limits are 0° and 360° , $\int_0^{2\pi}$); next one integrates with respect to α (the limits are 0° and 180° , $\int_0^\pi$, for, when the semicircle moves about its diameter, the spherical shell arises); finally one must, in order to obtain the infinitely many spherical shells, integrate with

respect to r , $\int_0^r$.

Thus one has, for the force with which the whole mass m attracts M ,

$$kp \iiint \frac{r^2 \sin \Theta \, dr \, d\Theta \, d\alpha (a - r \cos \alpha)}{(r^2 + a^2 - 2ra \cos \alpha)^{\frac{3}{2}}};$$

and herein a further proof that the infinite, at all points in the analysis, joins together with the finite, that the infinite is thus after all finite, and the finite infinite.